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Cho et al.

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(54) **LENS MOVING APPARATUS**

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G02B 7/09 (2021.01)

(Continued)

(52) **U.S. Cl.**

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(Continued)

(58) **Field of Classification Search**

CPC . G03B 3/10; G03B 5/00; G03B 17/02; G03B 5/04; G03B 13/36; G03B 17/12;

(Continued)

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Primary Examiner — Ephrem Z Mebrahtu

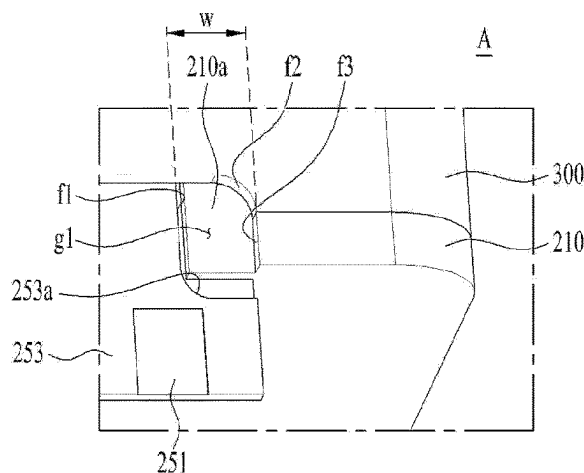
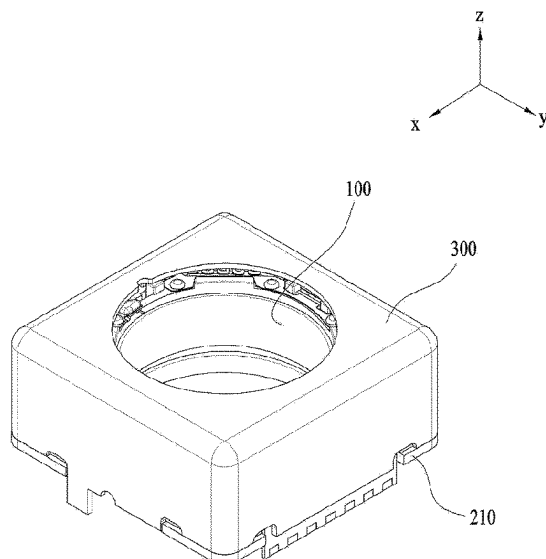
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(57)

ABSTRACT

A lens moving apparatus includes a bobbin on which a first coil is disposed, a first magnet disposed around the bobbin to face the first coil, and a housing, which is disposed to surround at least a portion of the bobbin and has a first magnet mounting seat, which receives the first magnet, wherein the housing is provided with an adhesive inlet, which allows a side portion of the first magnet mounting seat to communicate with an outside surface of the housing.

20 Claims, 27 Drawing Sheets



Related U.S. Application Data

continuation of application No. 14/884,412, filed on Oct. 15, 2015, now Pat. No. 10,451,891.

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G03B 5/00 (2021.01)
- (52) **U.S. Cl.**
 CPC *G03B 2205/0007* (2013.01); *G03B 2205/0069* (2013.01)
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- See application file for complete search history.

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FIG. 1

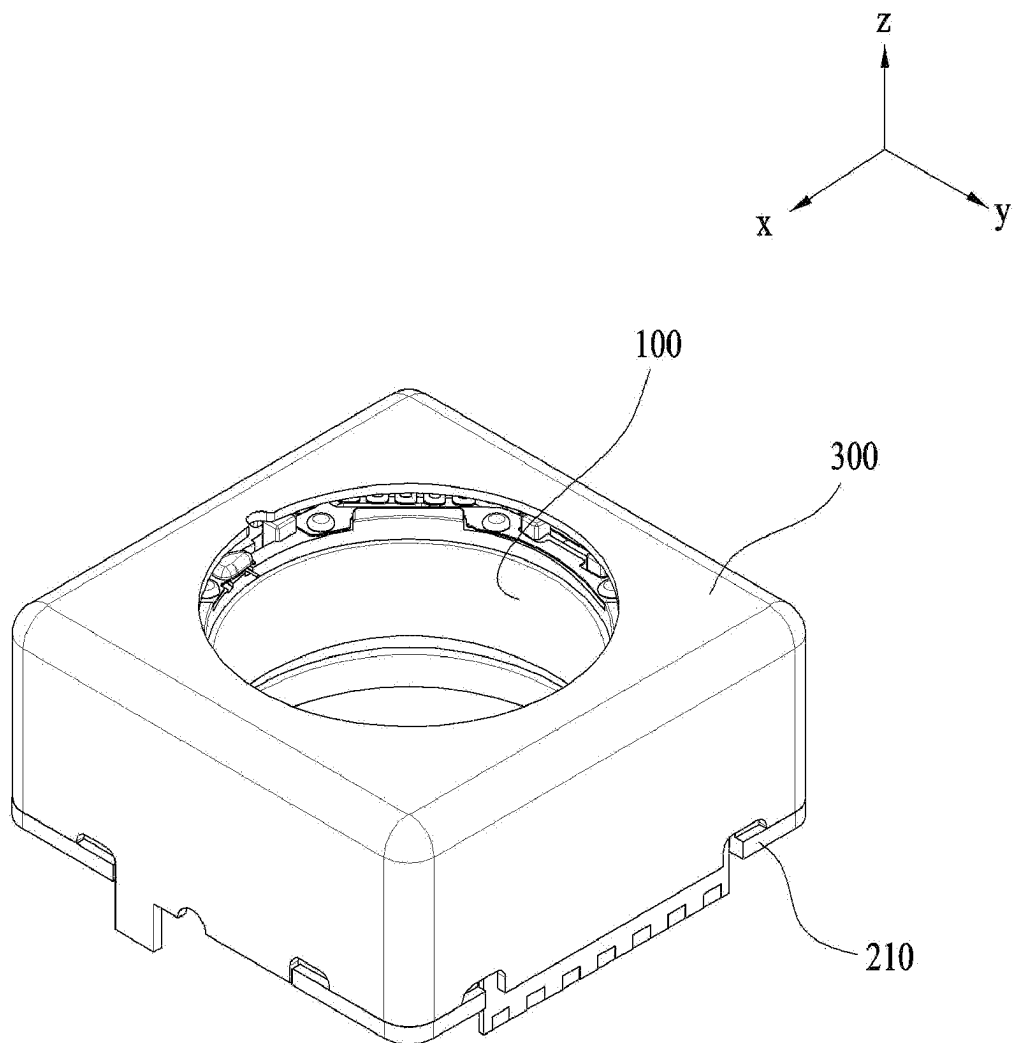


FIG. 2

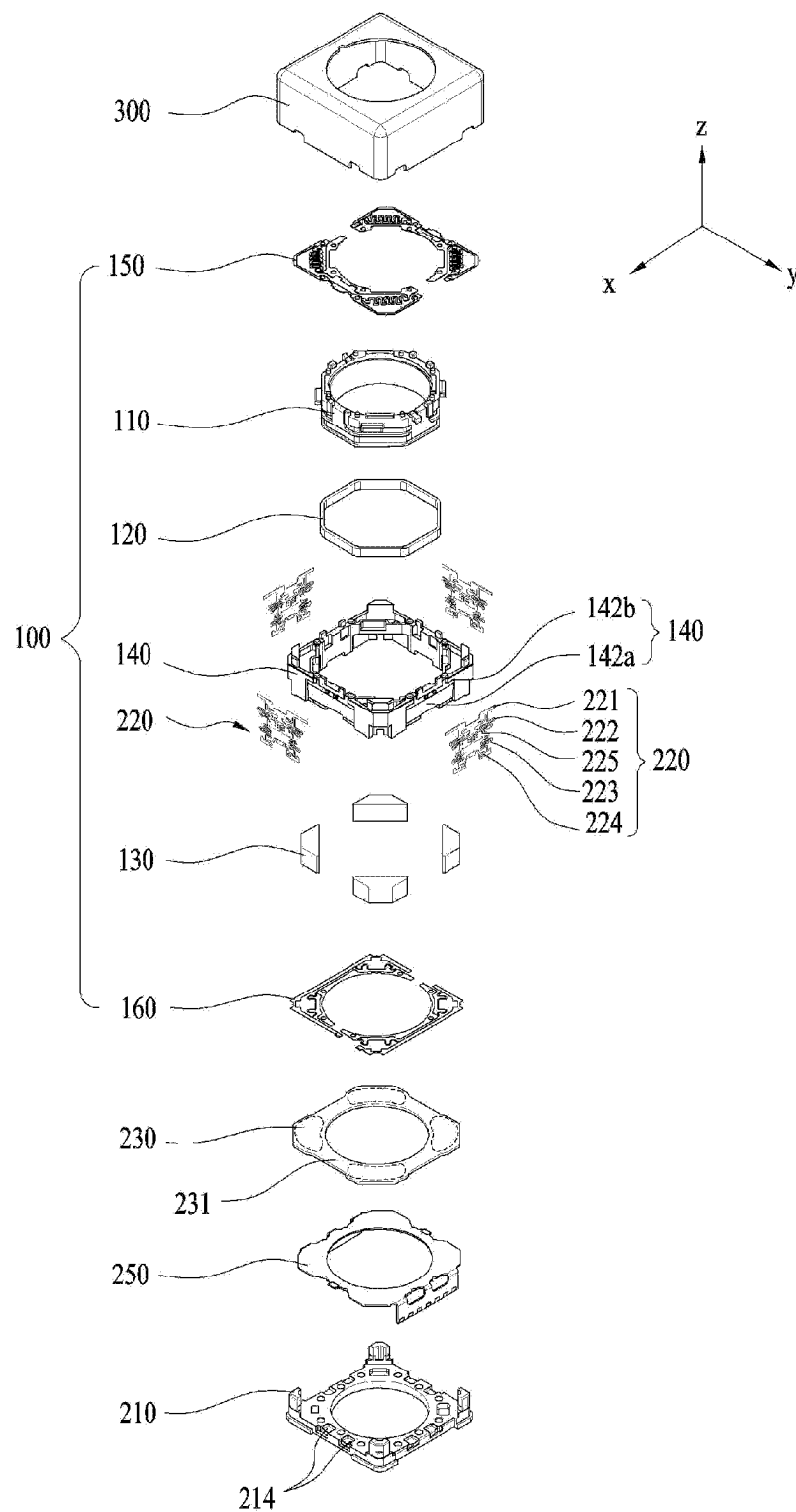


FIG. 3

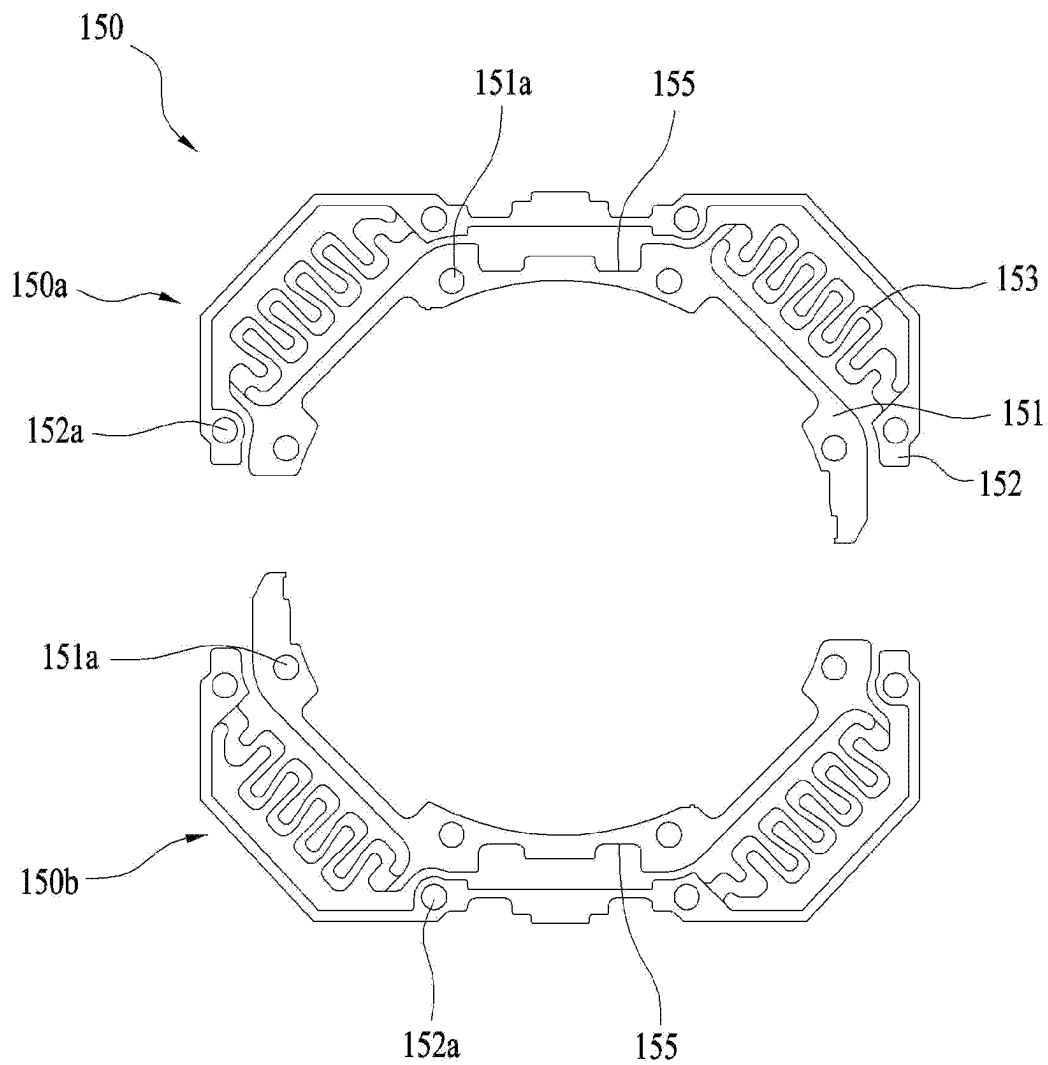


FIG. 4

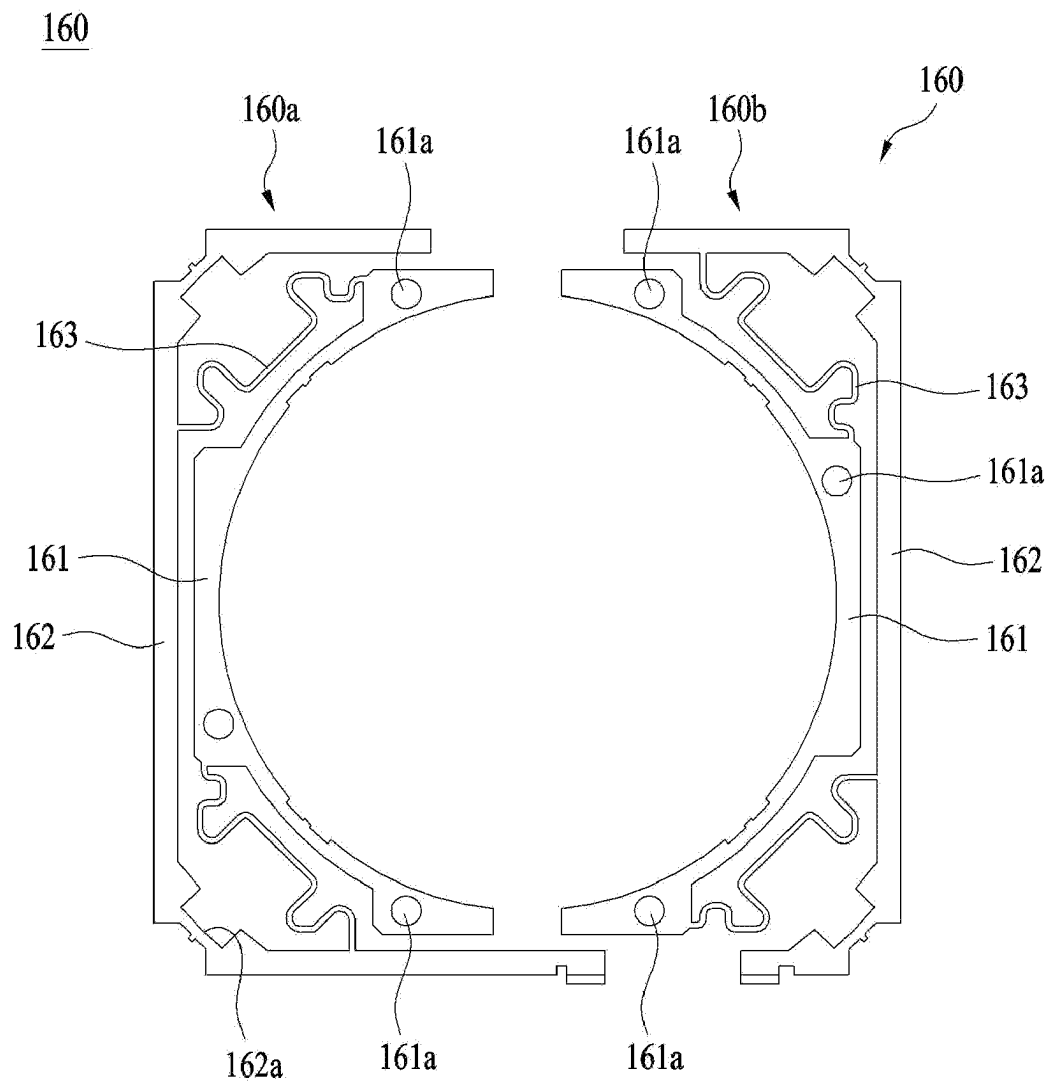


FIG. 5

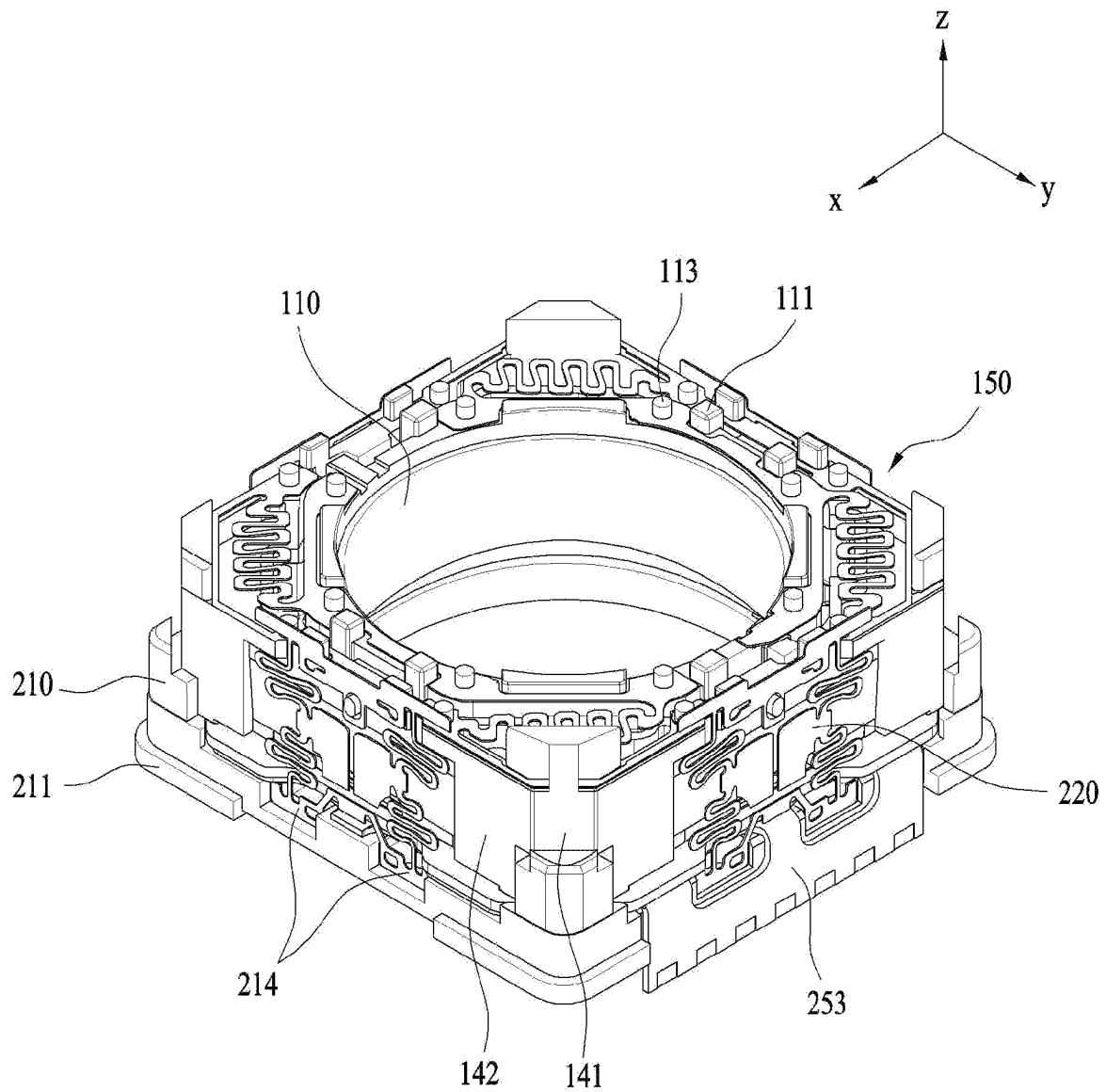


FIG. 6

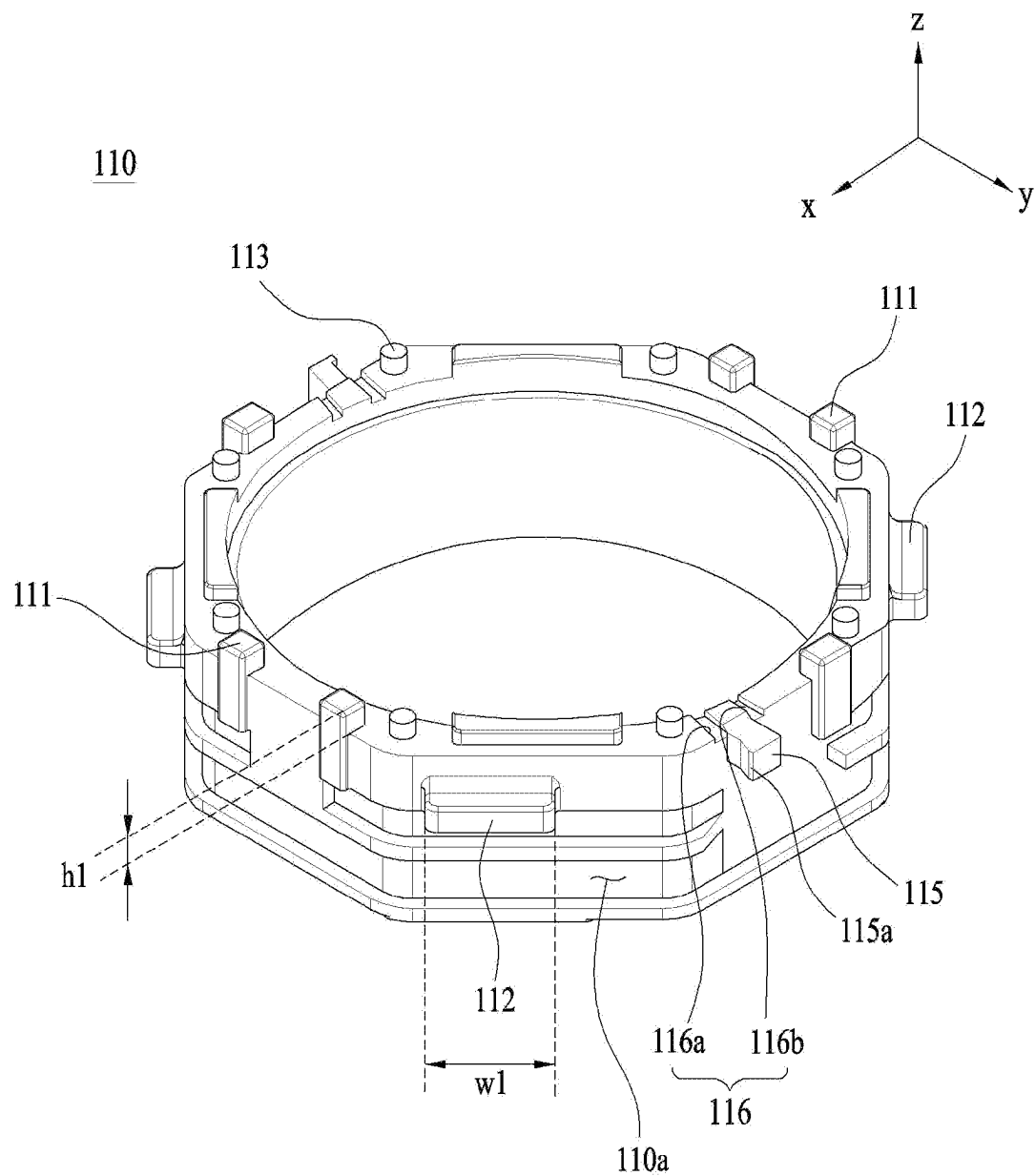


FIG. 7

220

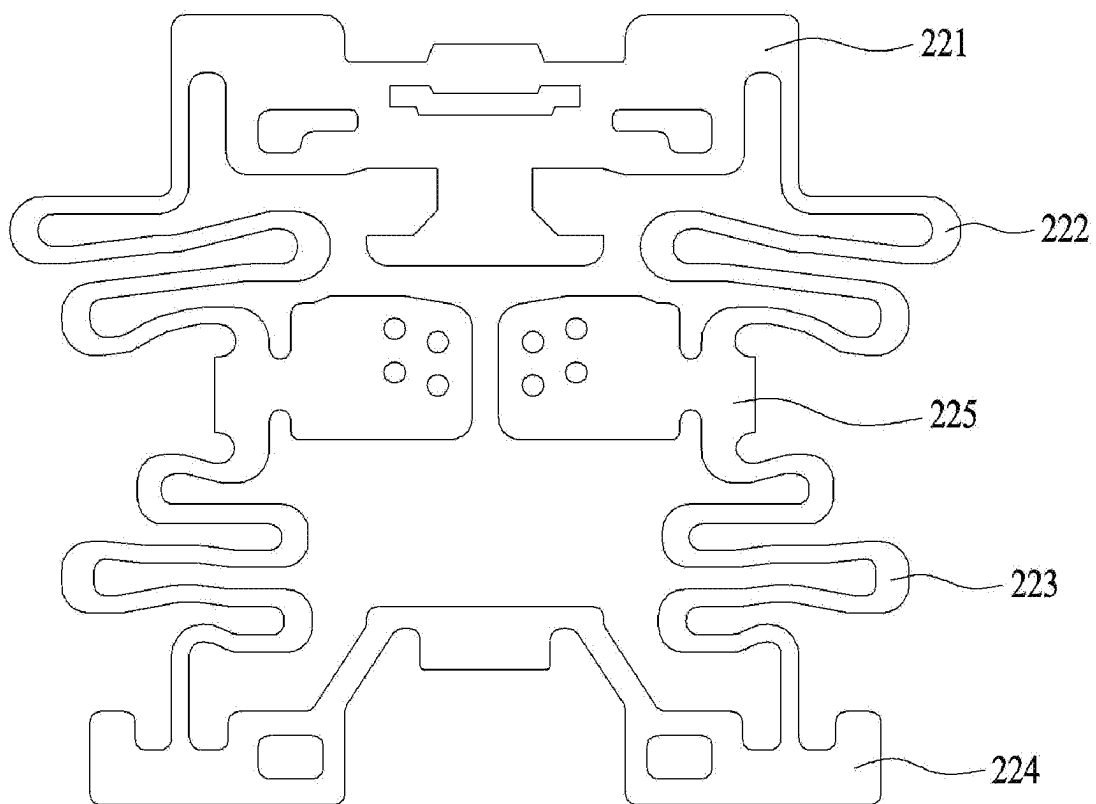


FIG. 8

220

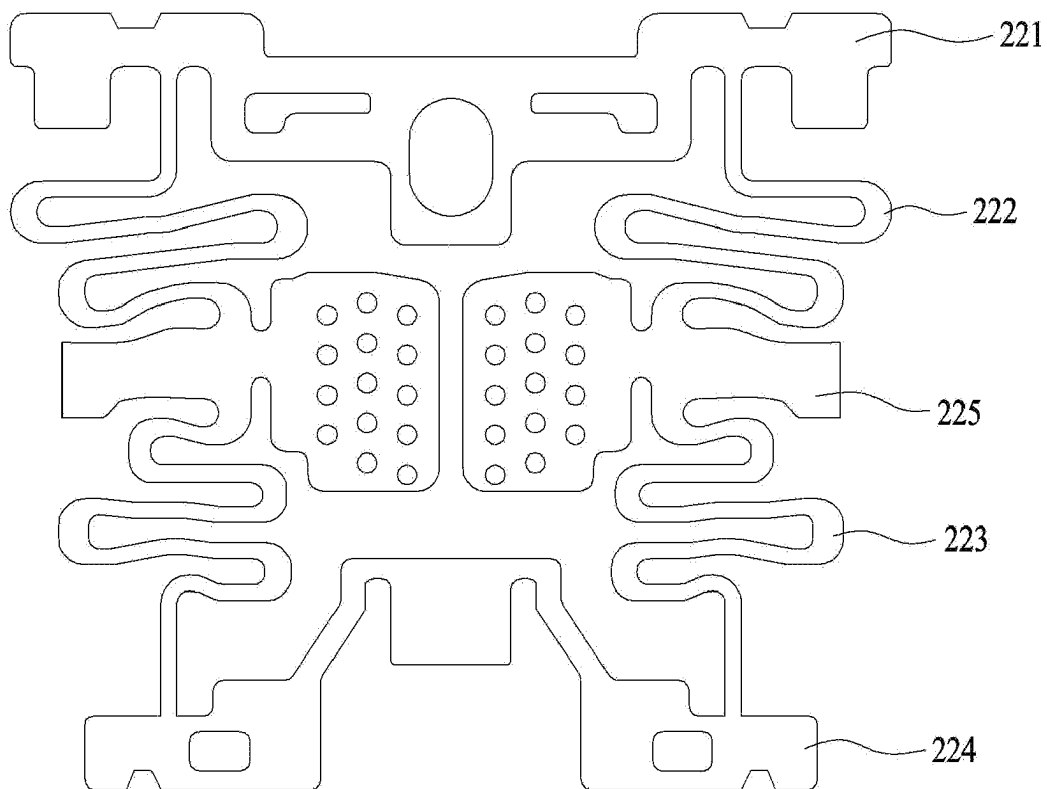


FIG. 9

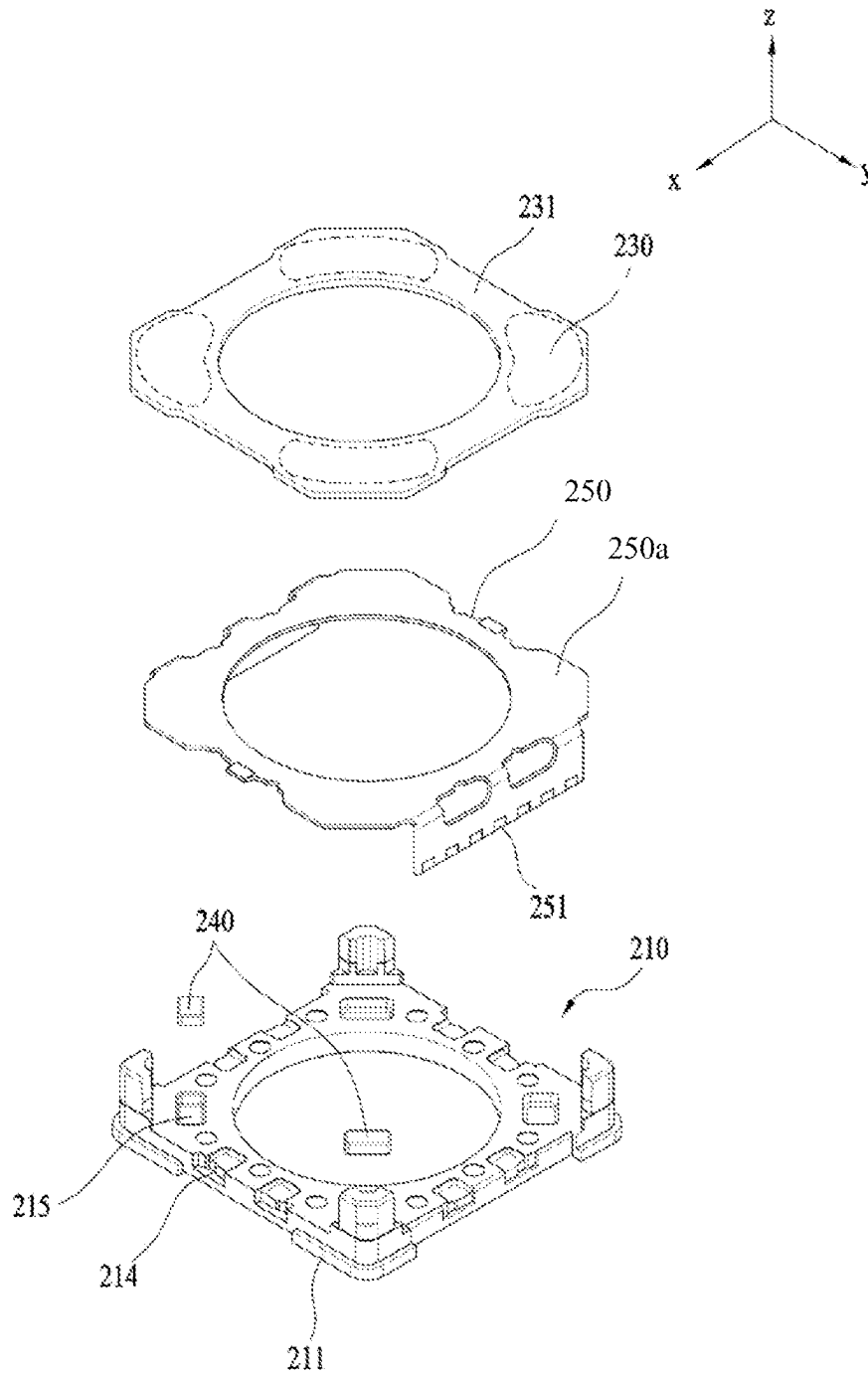


FIG. 10

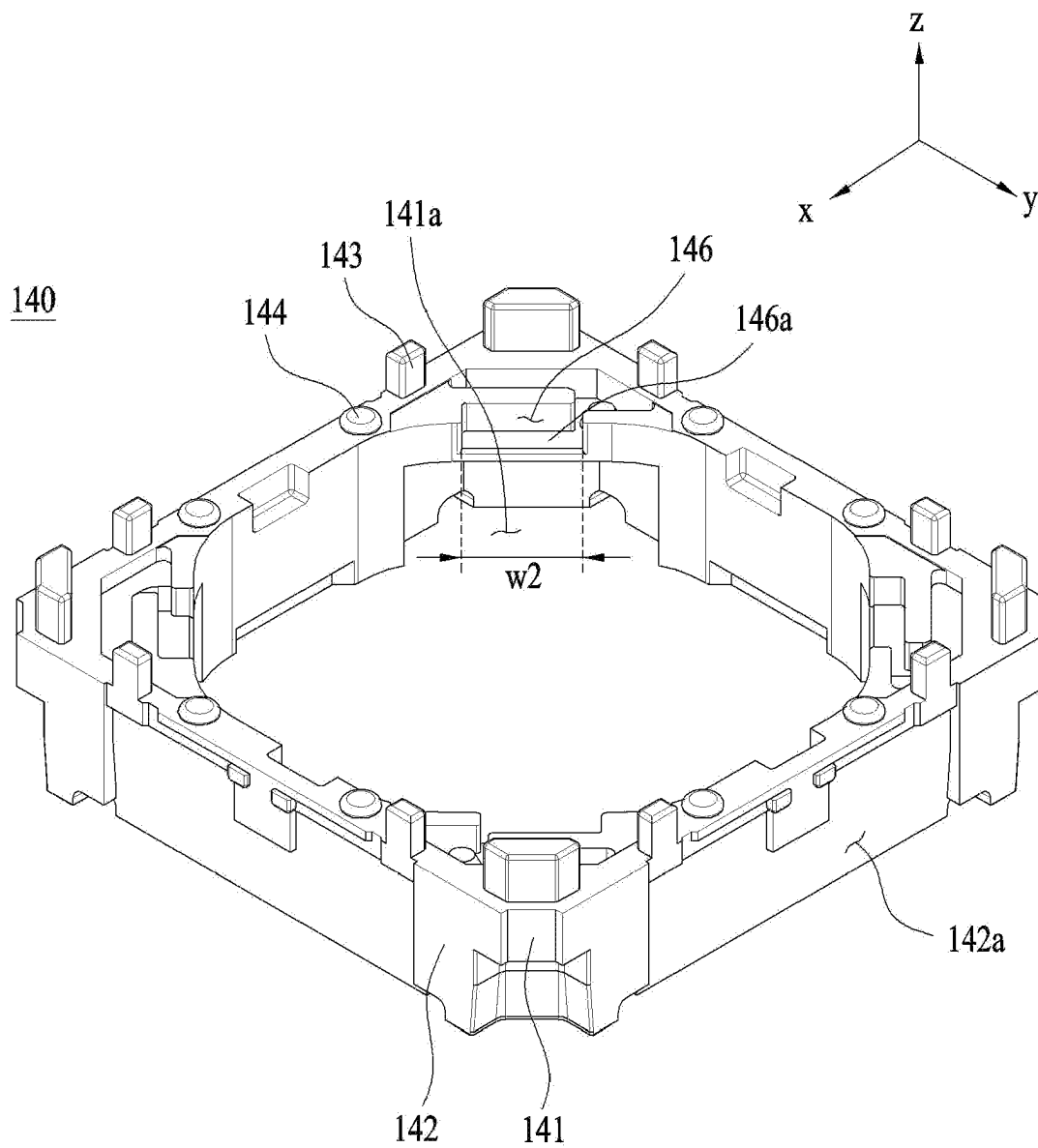


FIG. 11

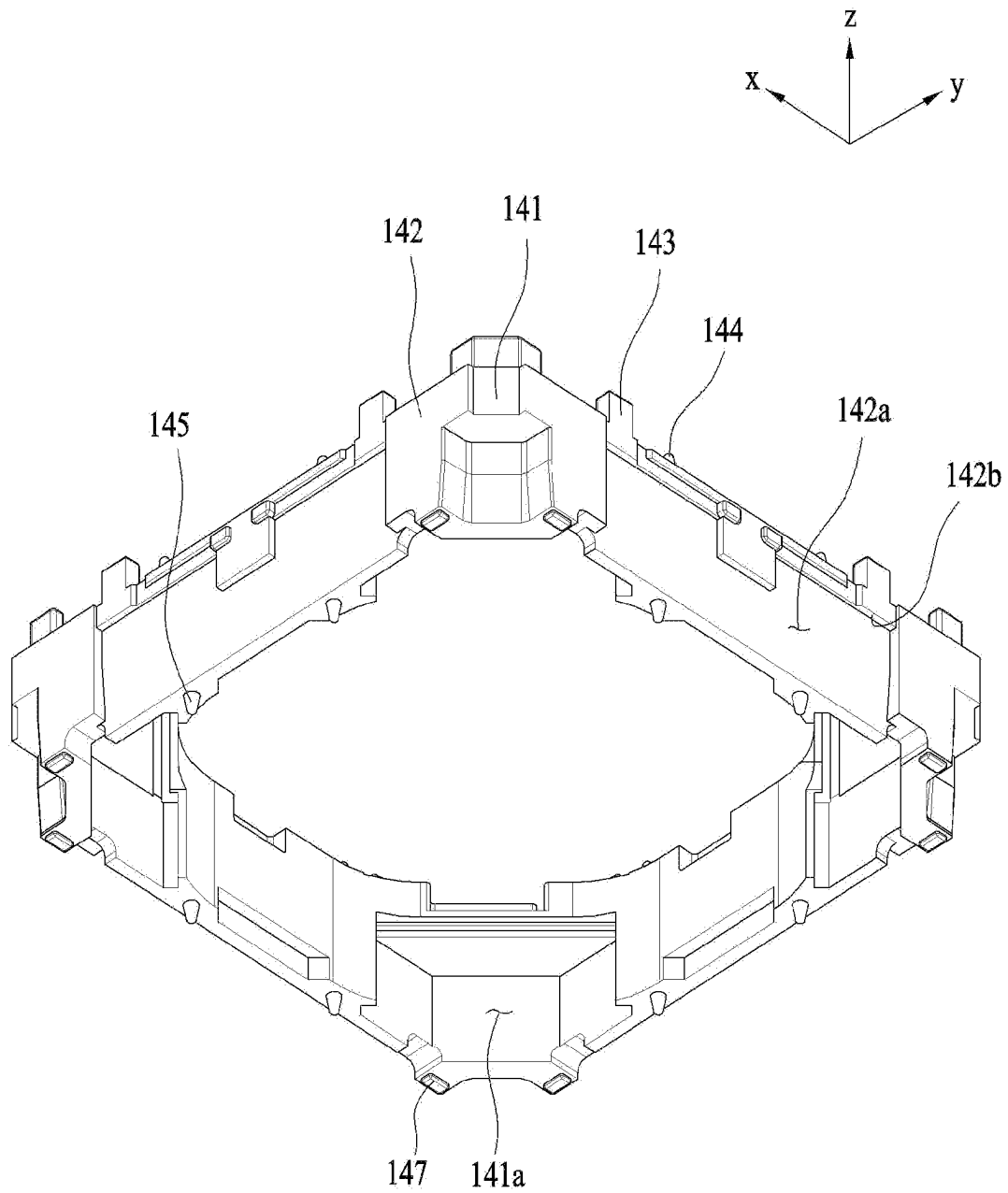


FIG. 12

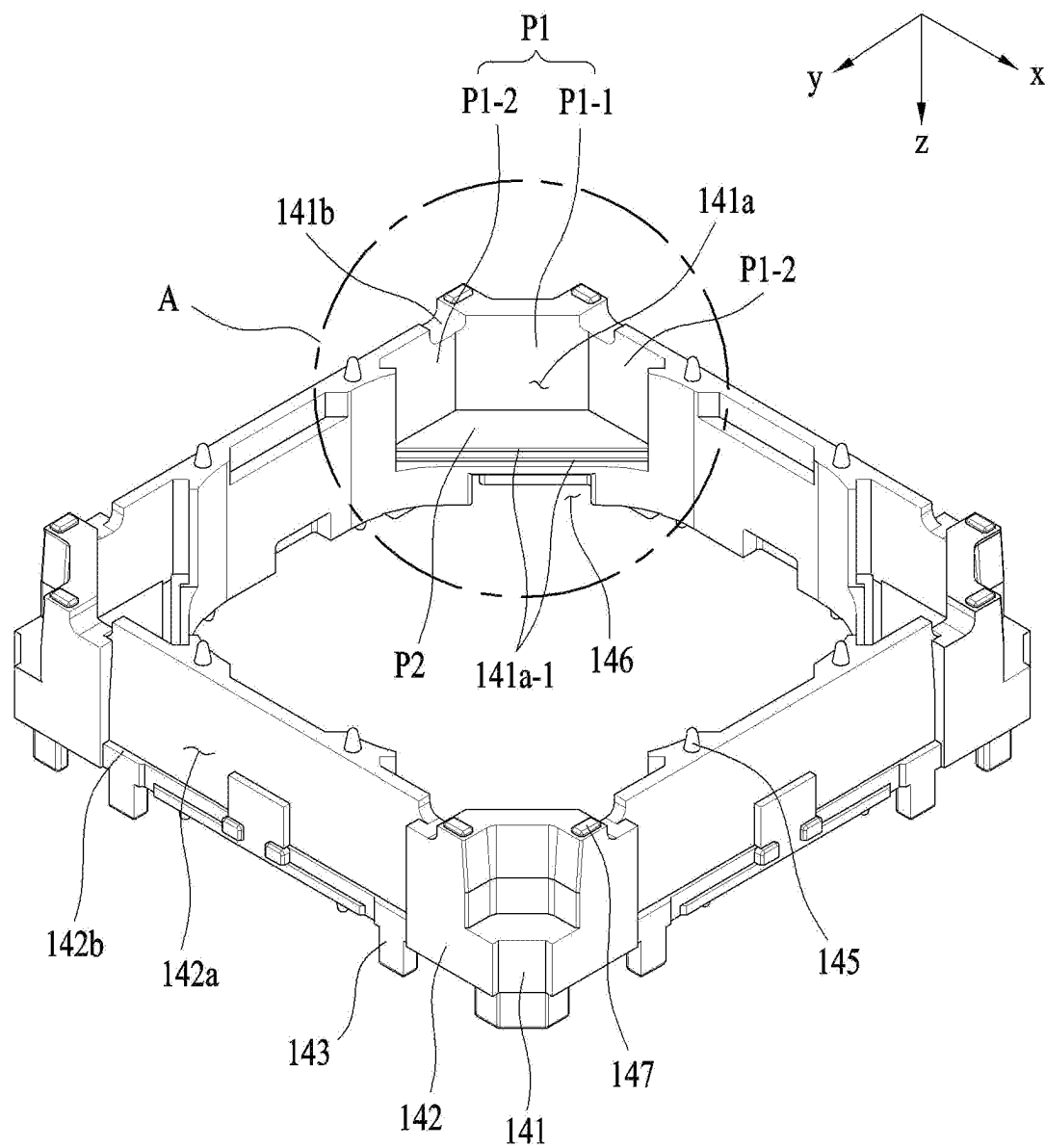


FIG. 13

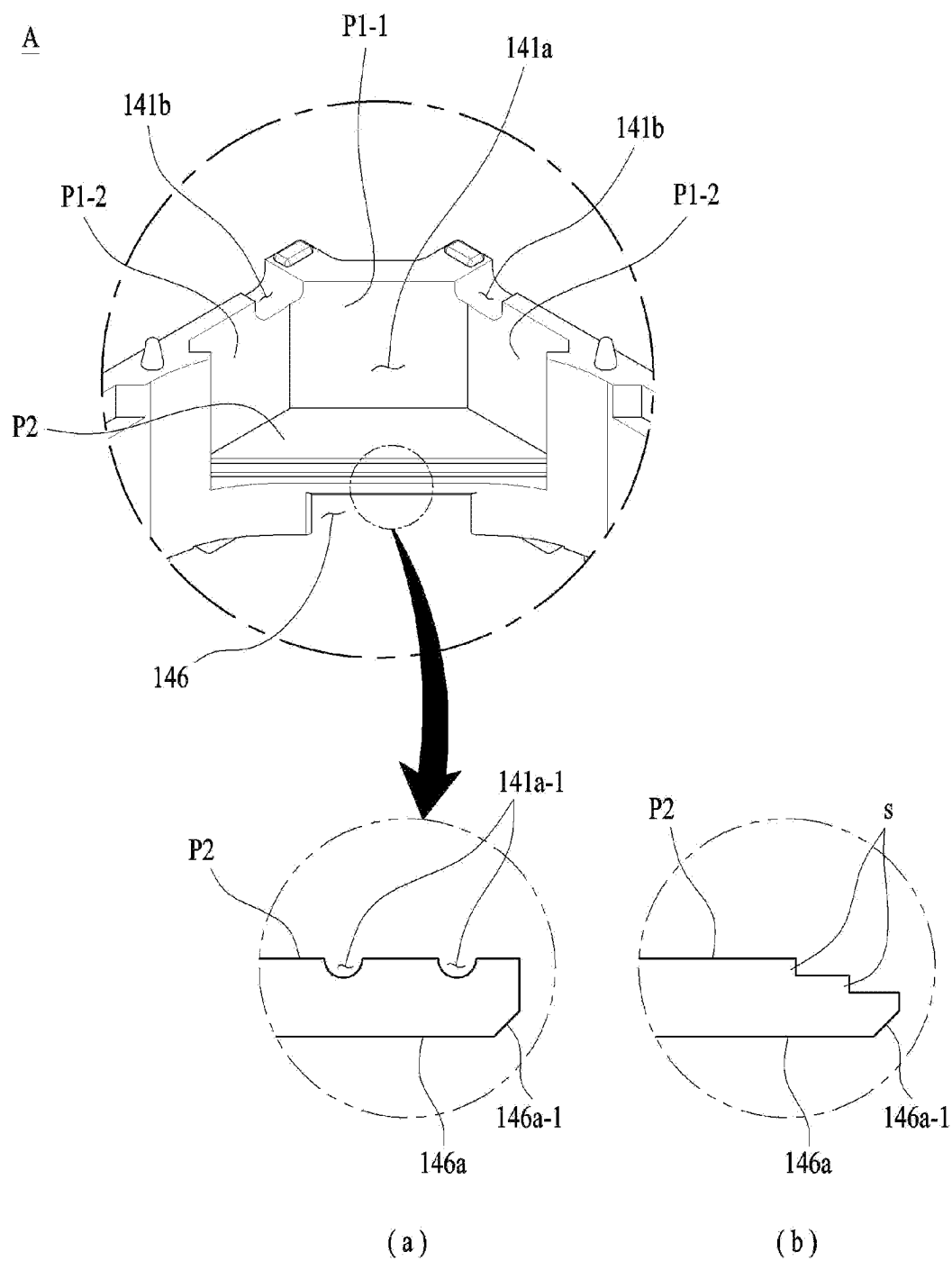


FIG. 14

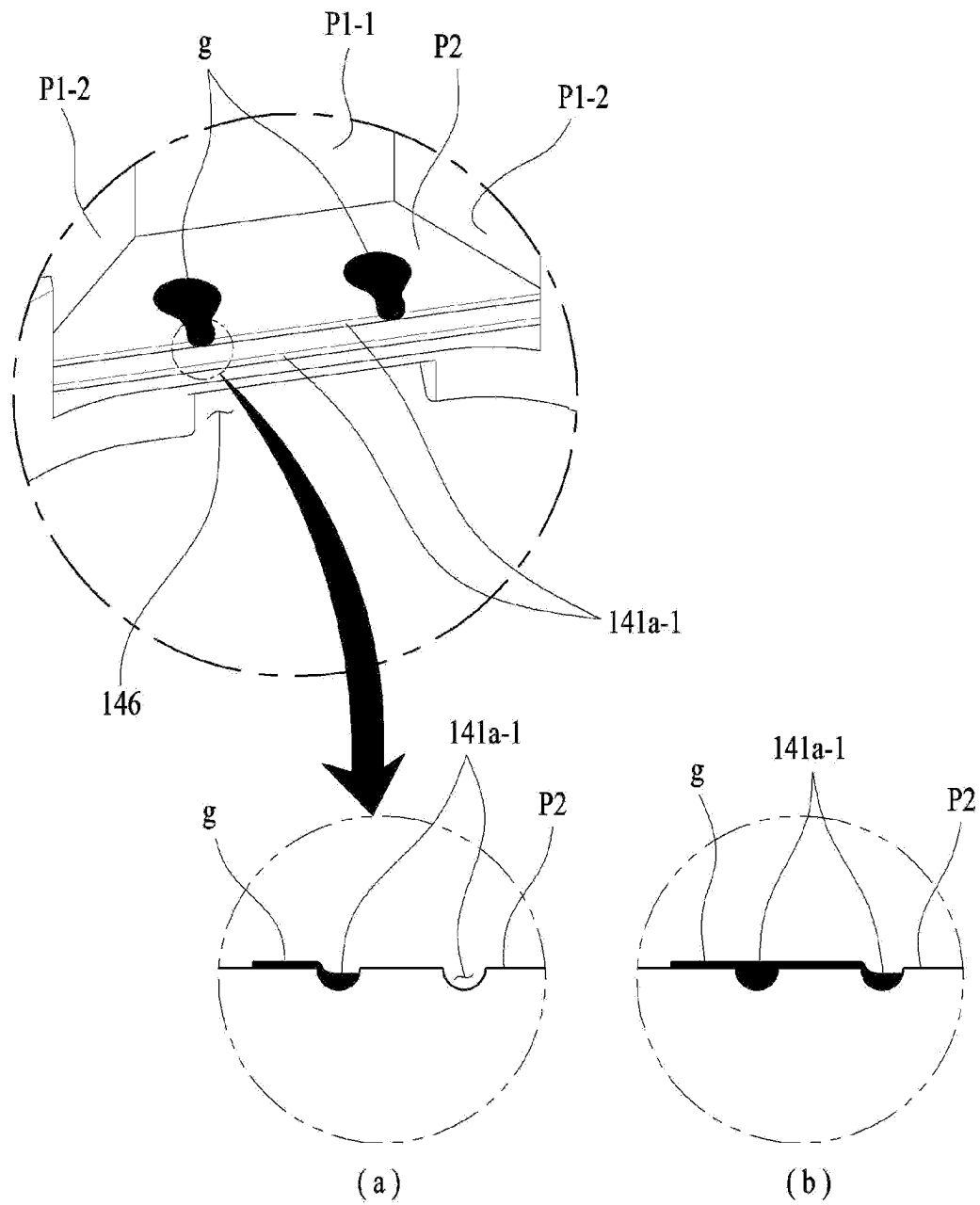


FIG. 15

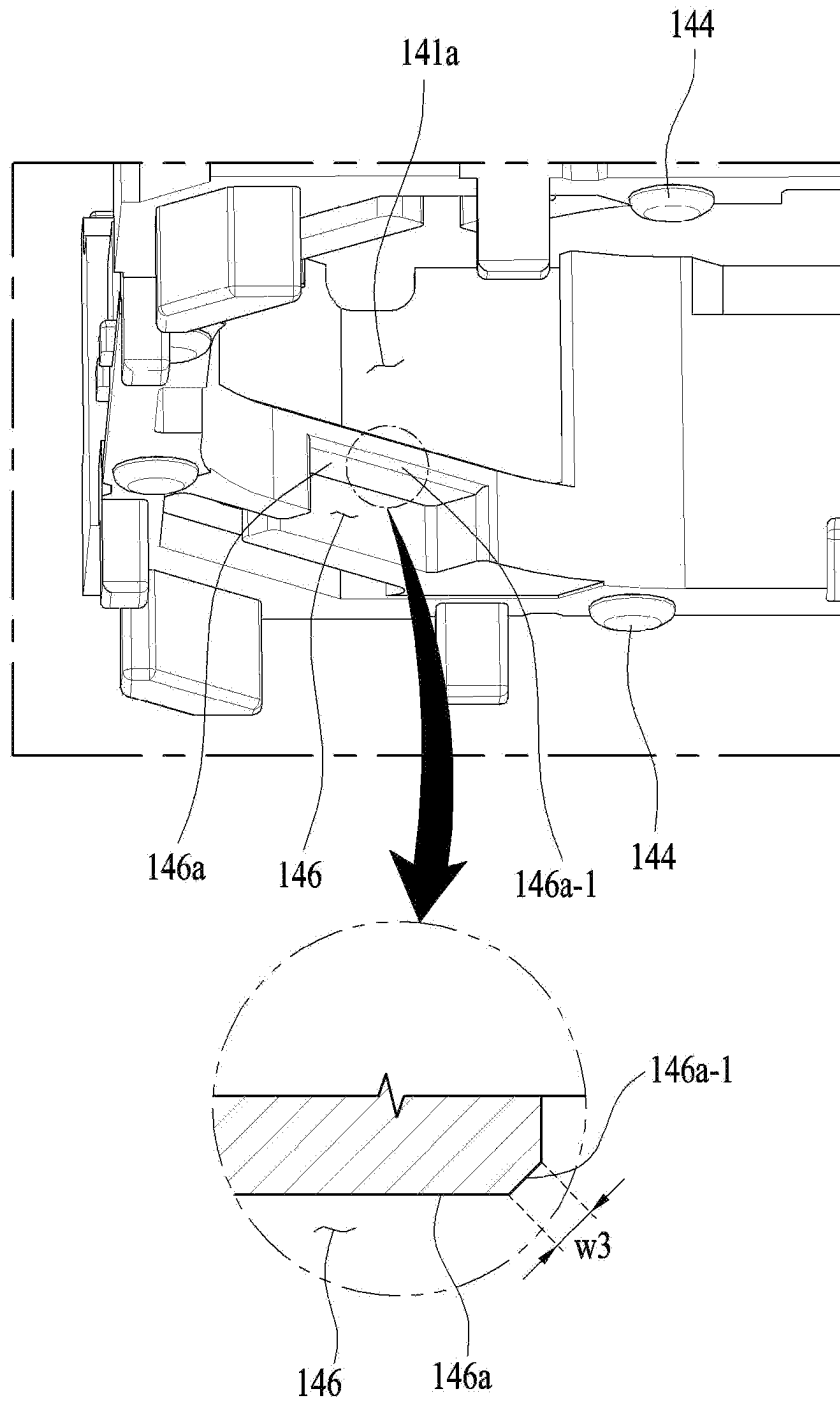


FIG. 16

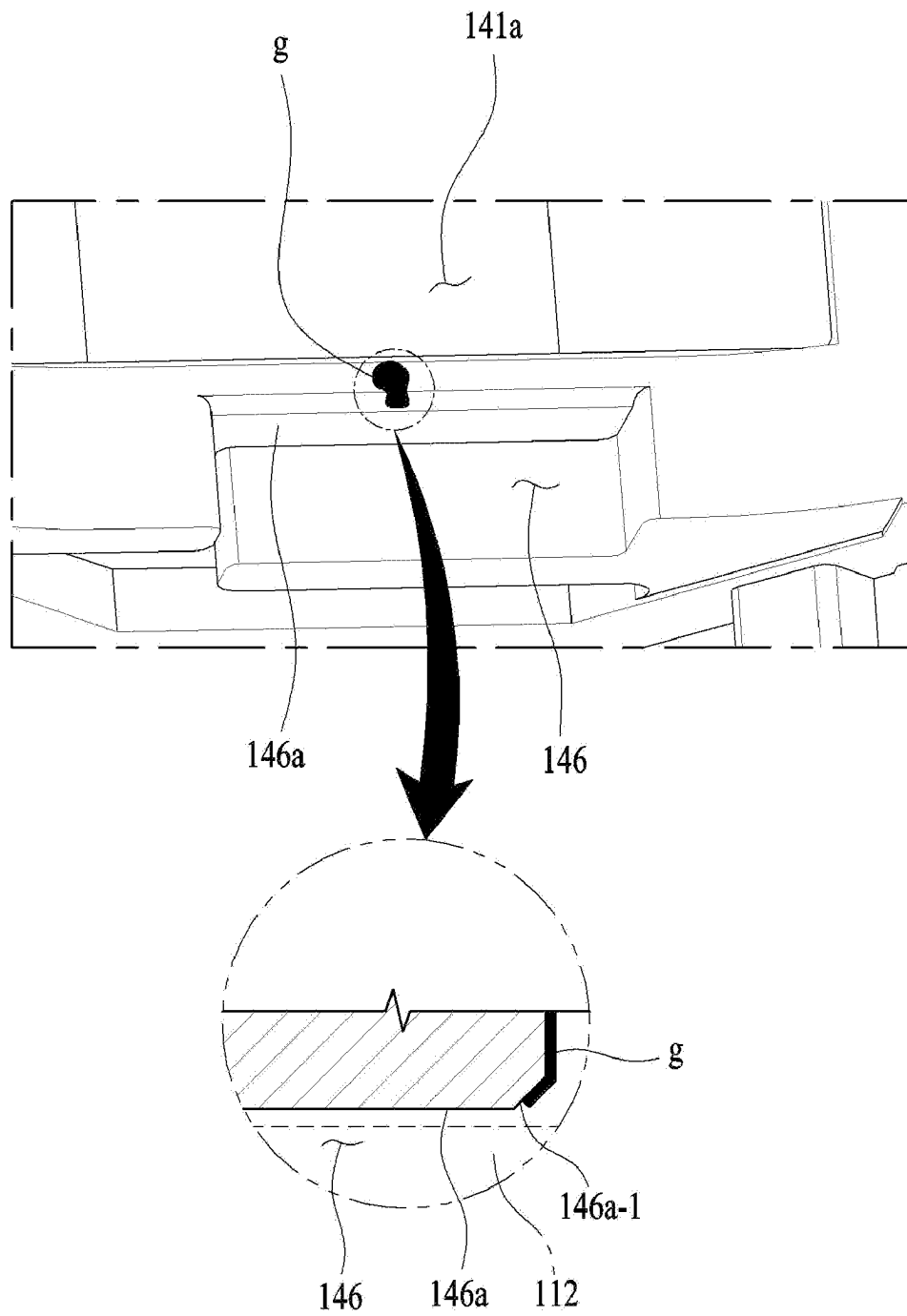


FIG. 17

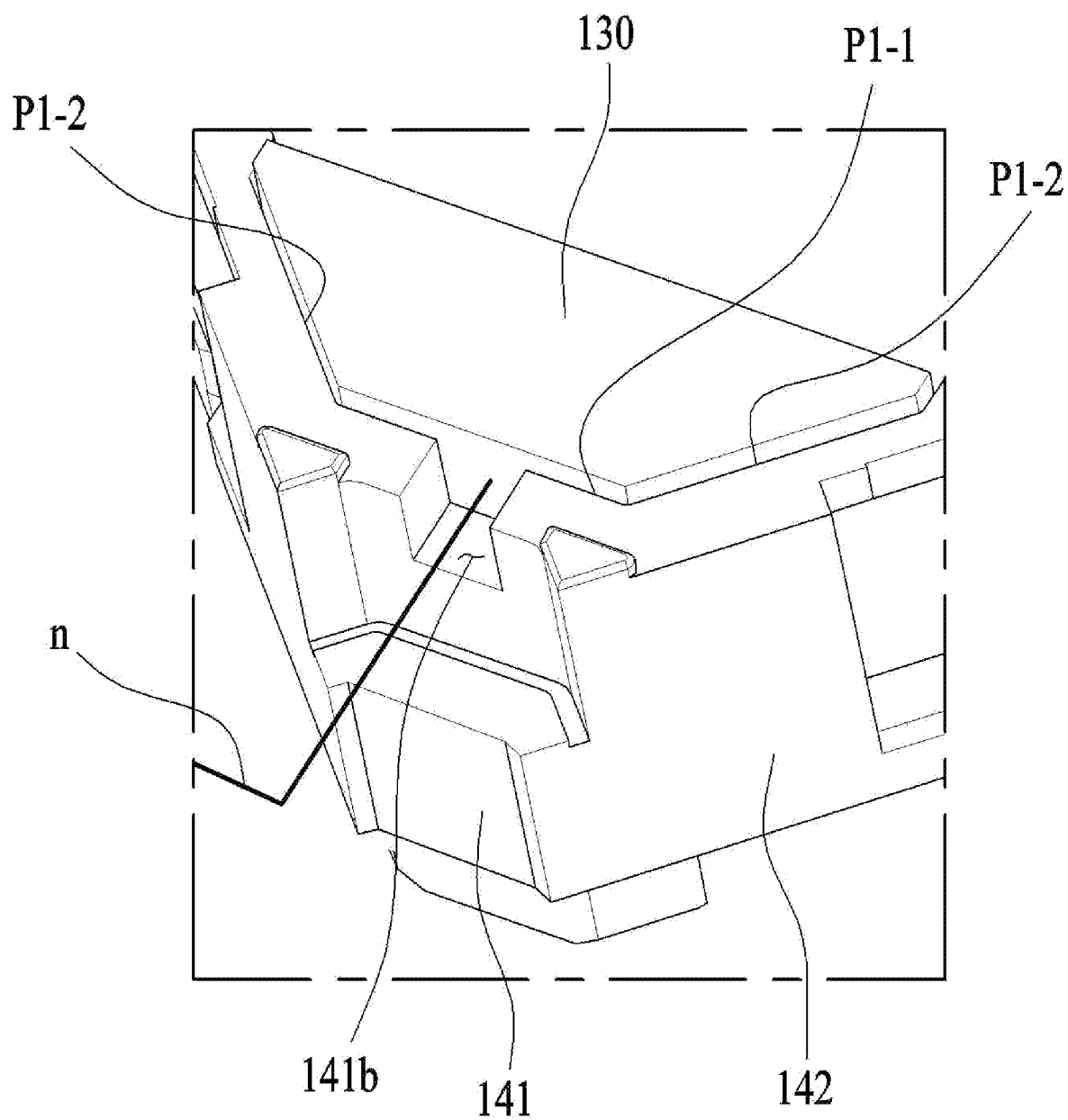


FIG. 18

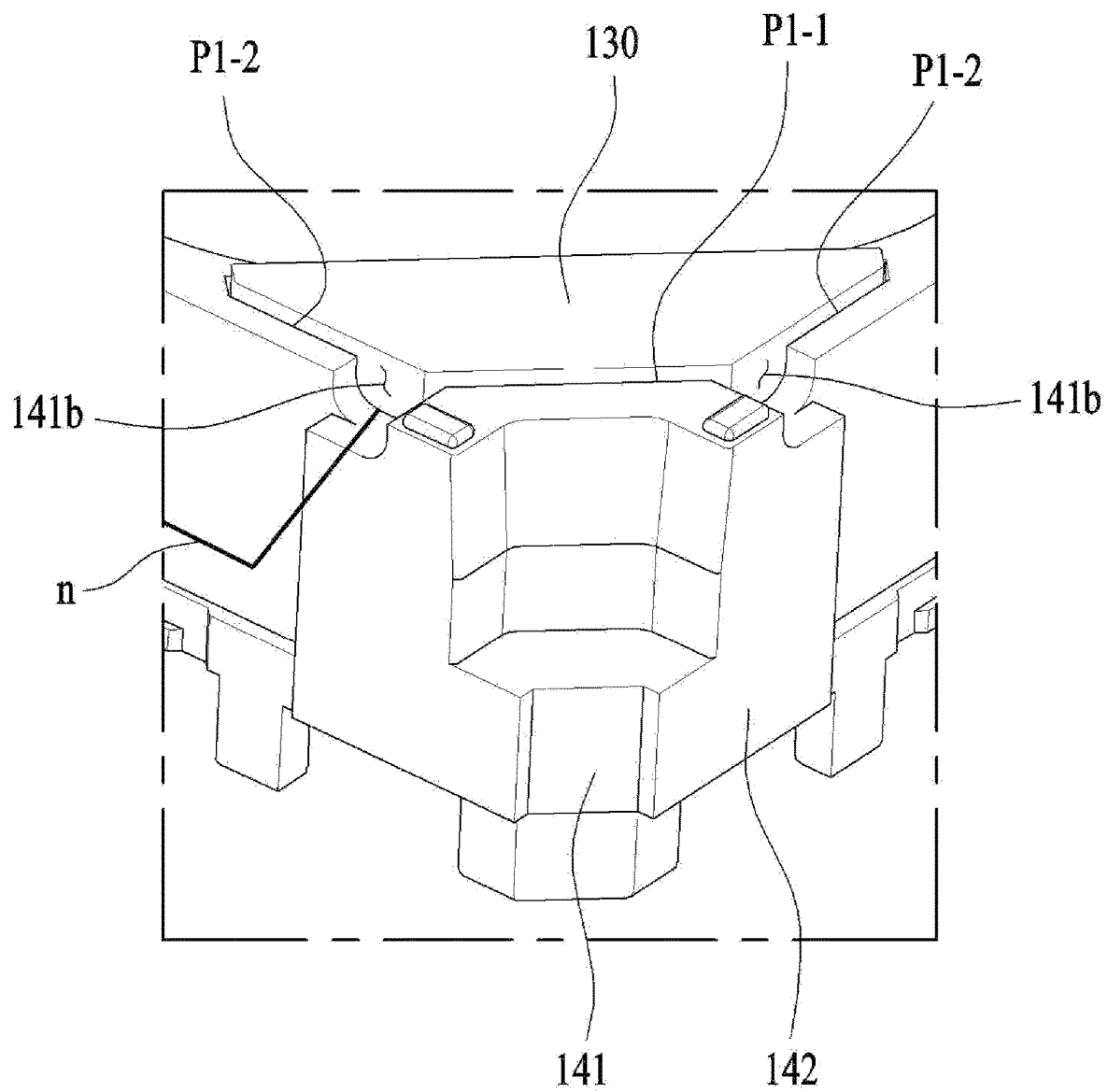


FIG. 19

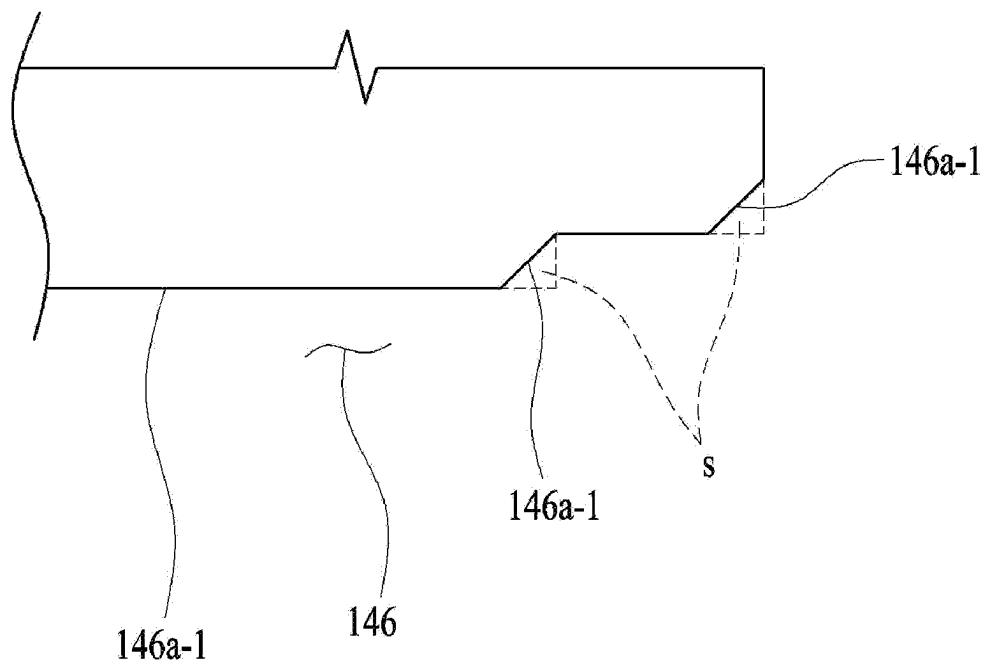


FIG. 20

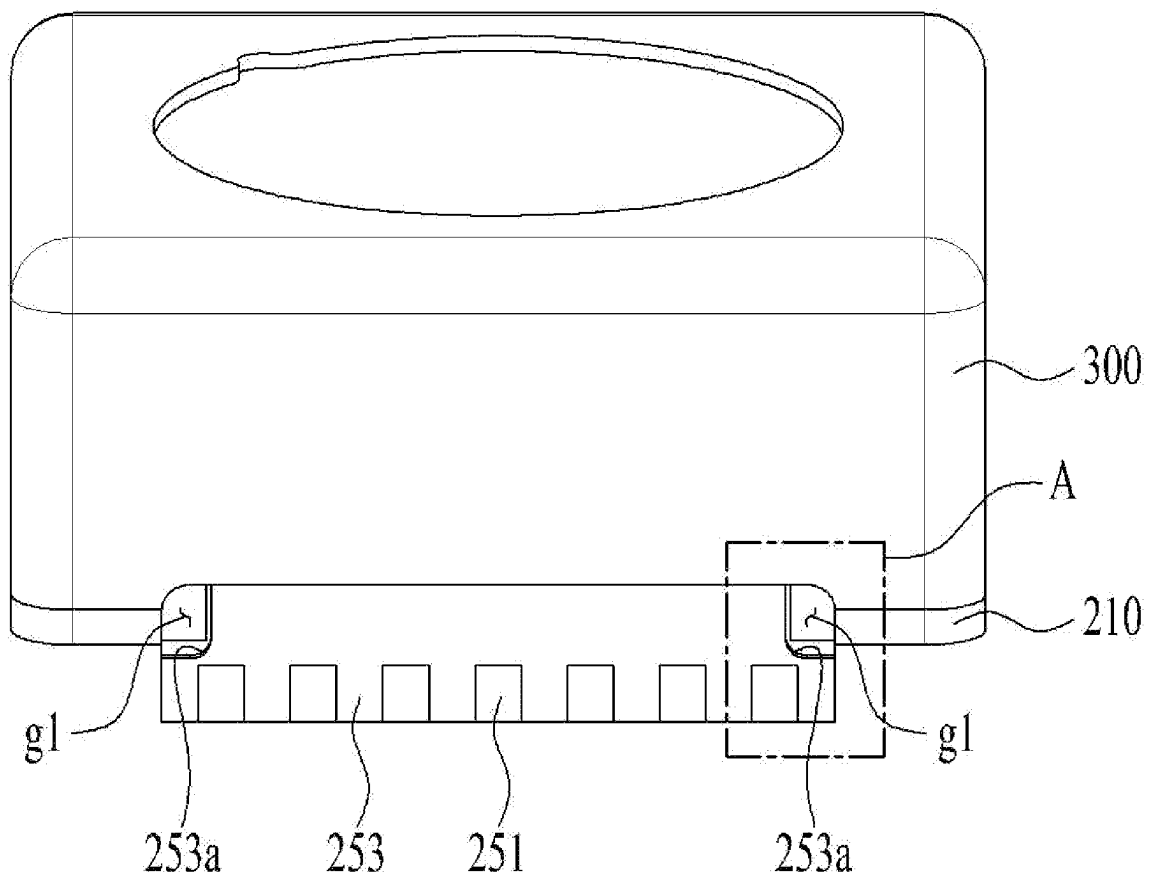


FIG. 21

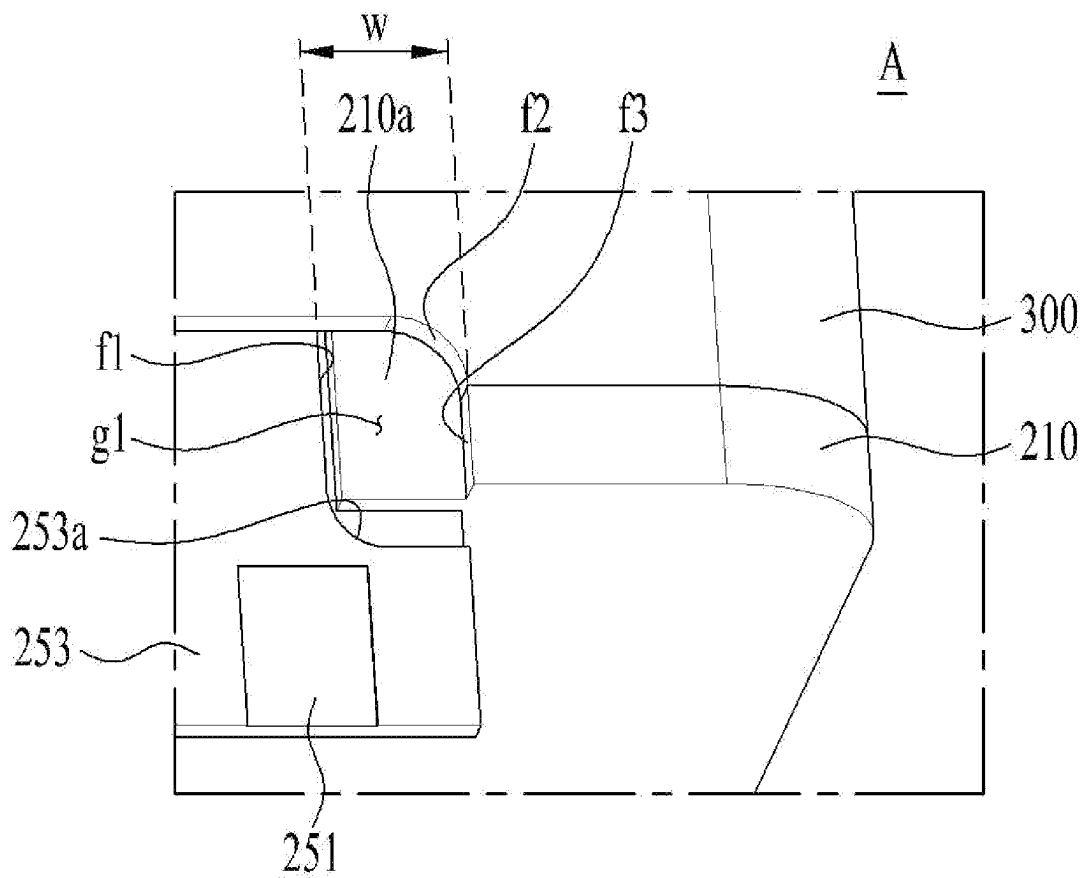


FIG. 22

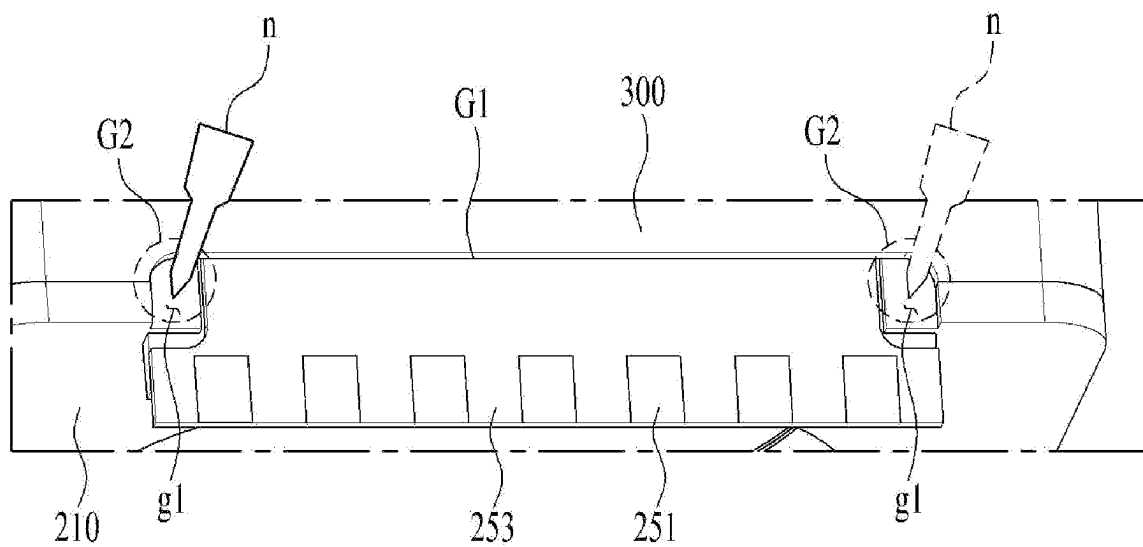


FIG. 23

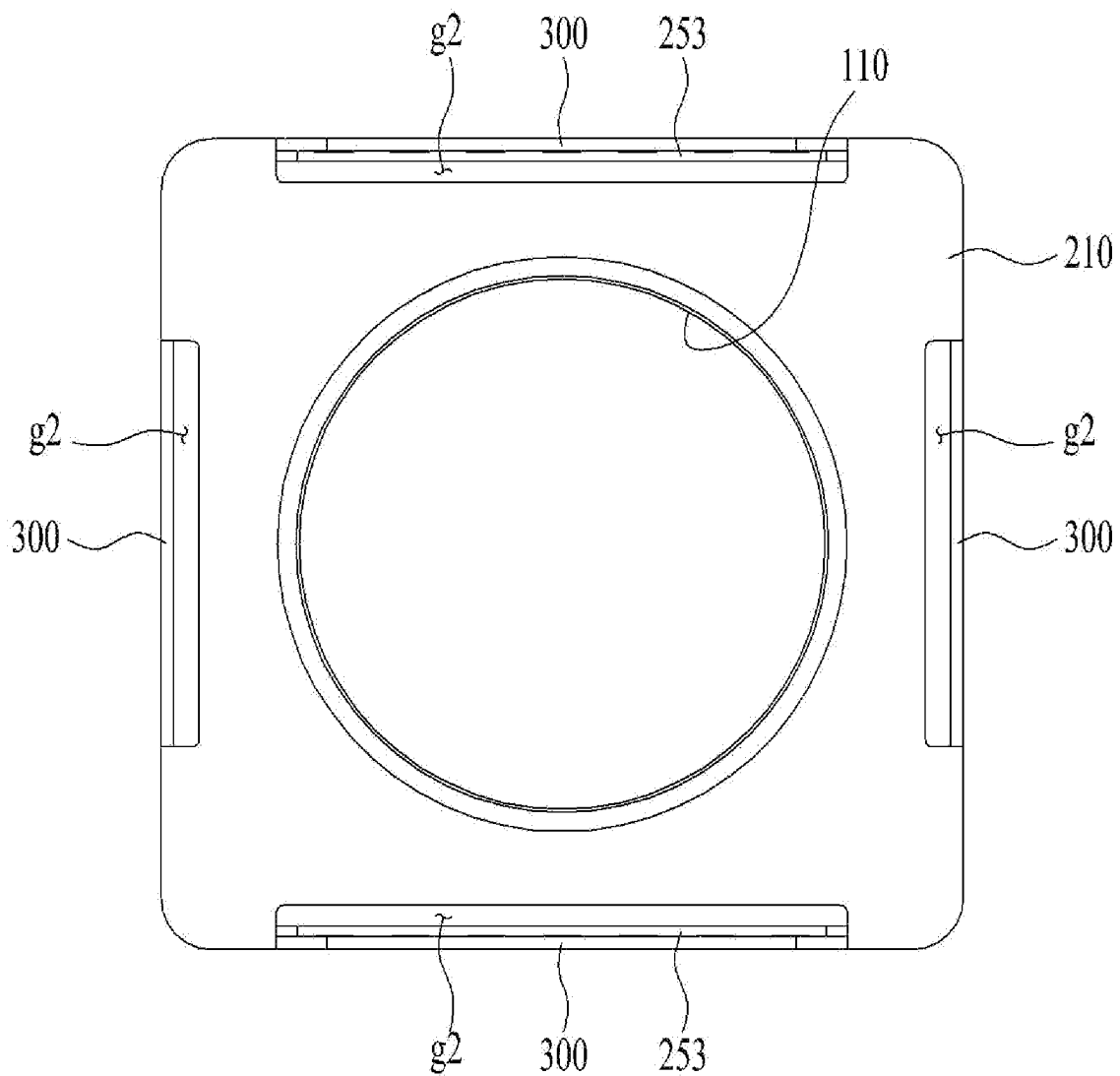
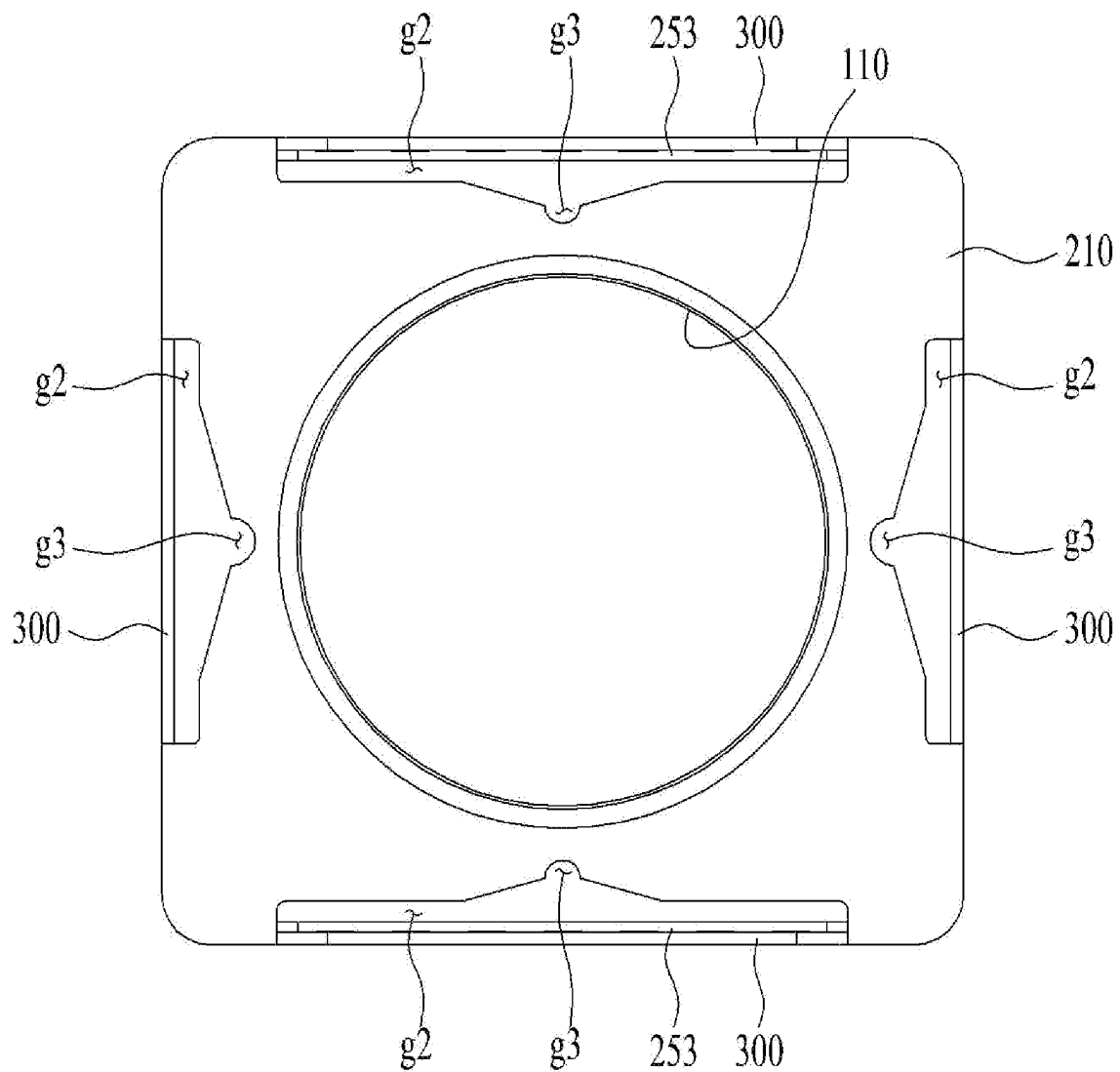


FIG. 24



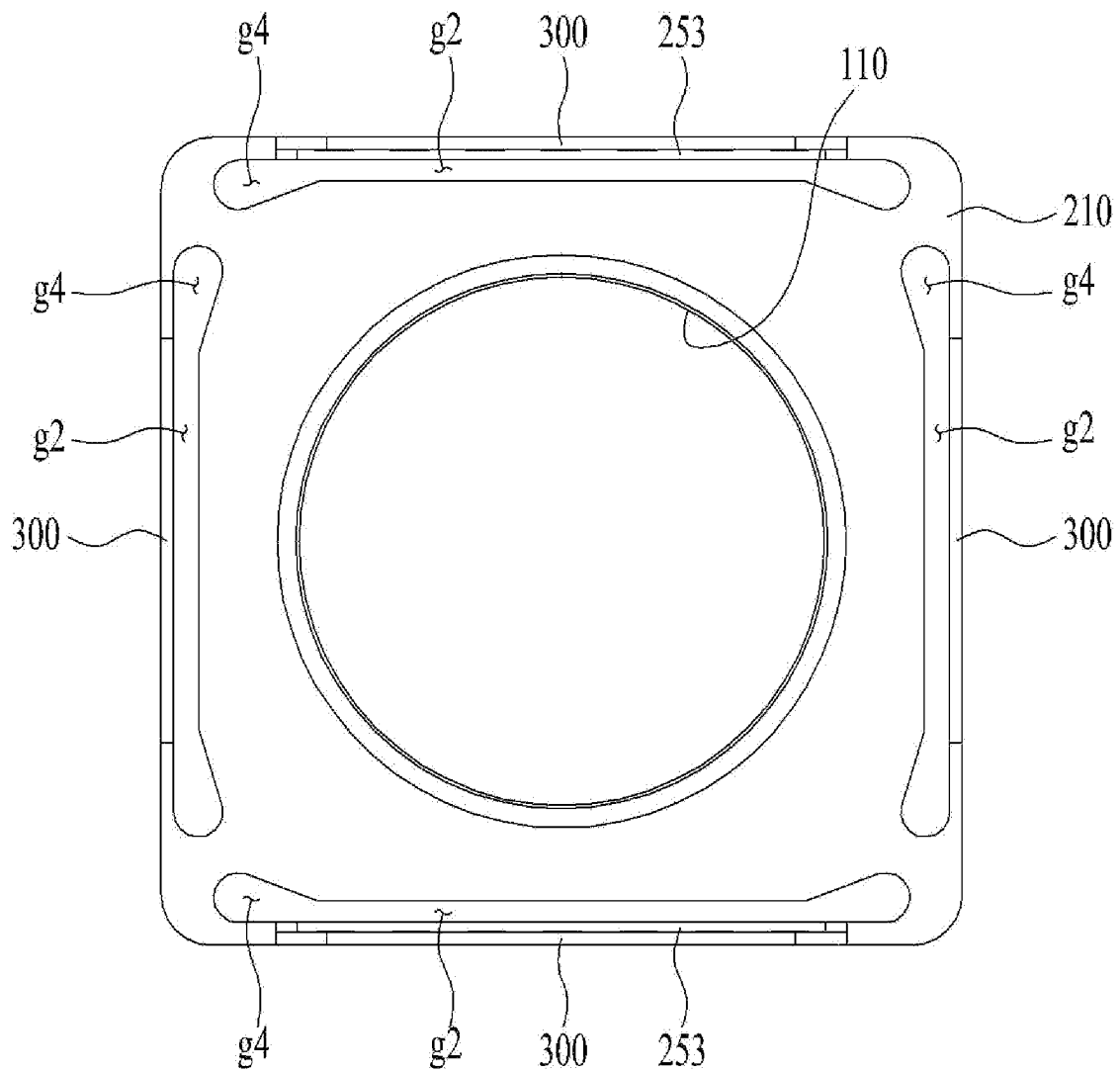


FIG. 26

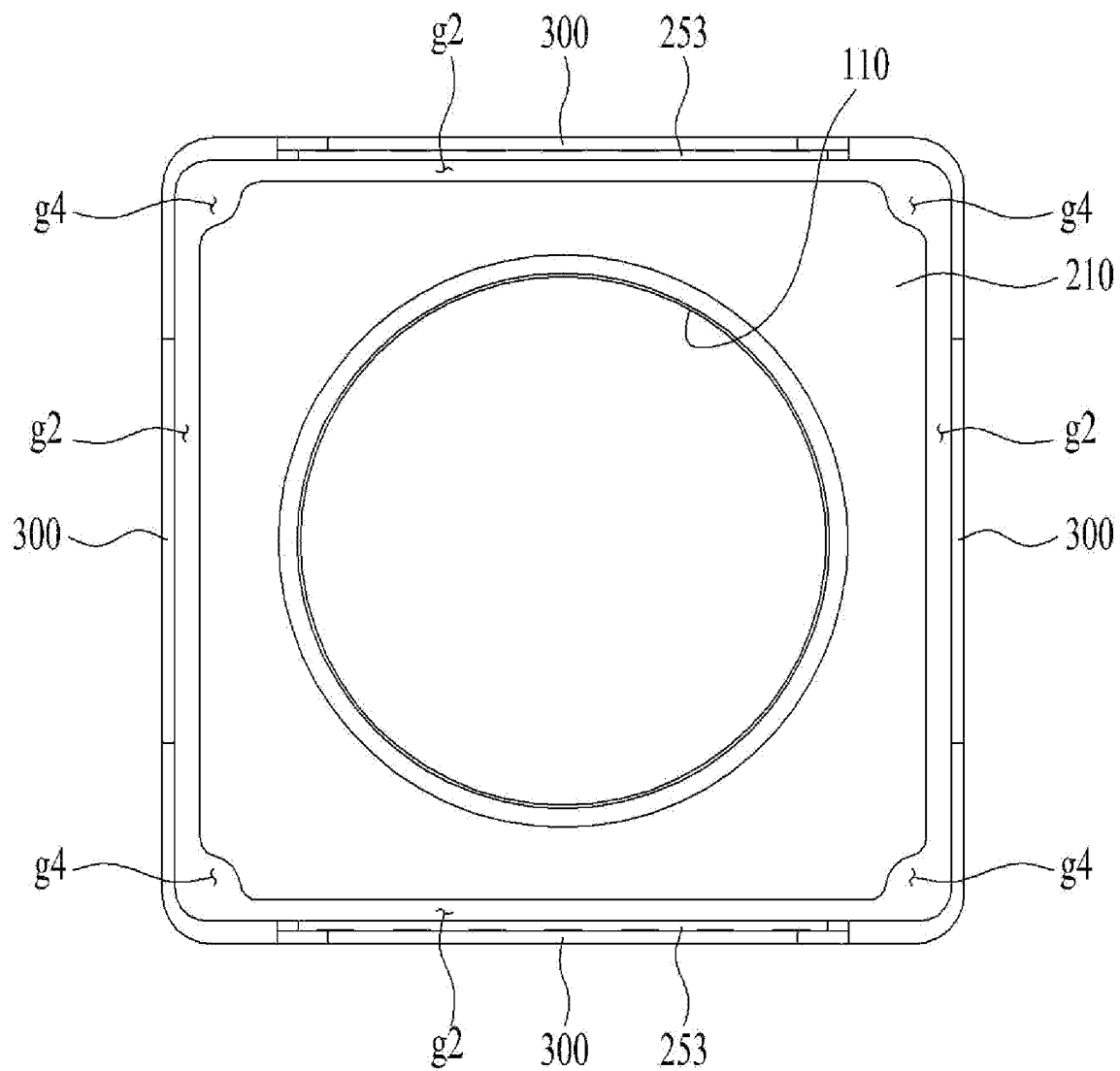


FIG. 27

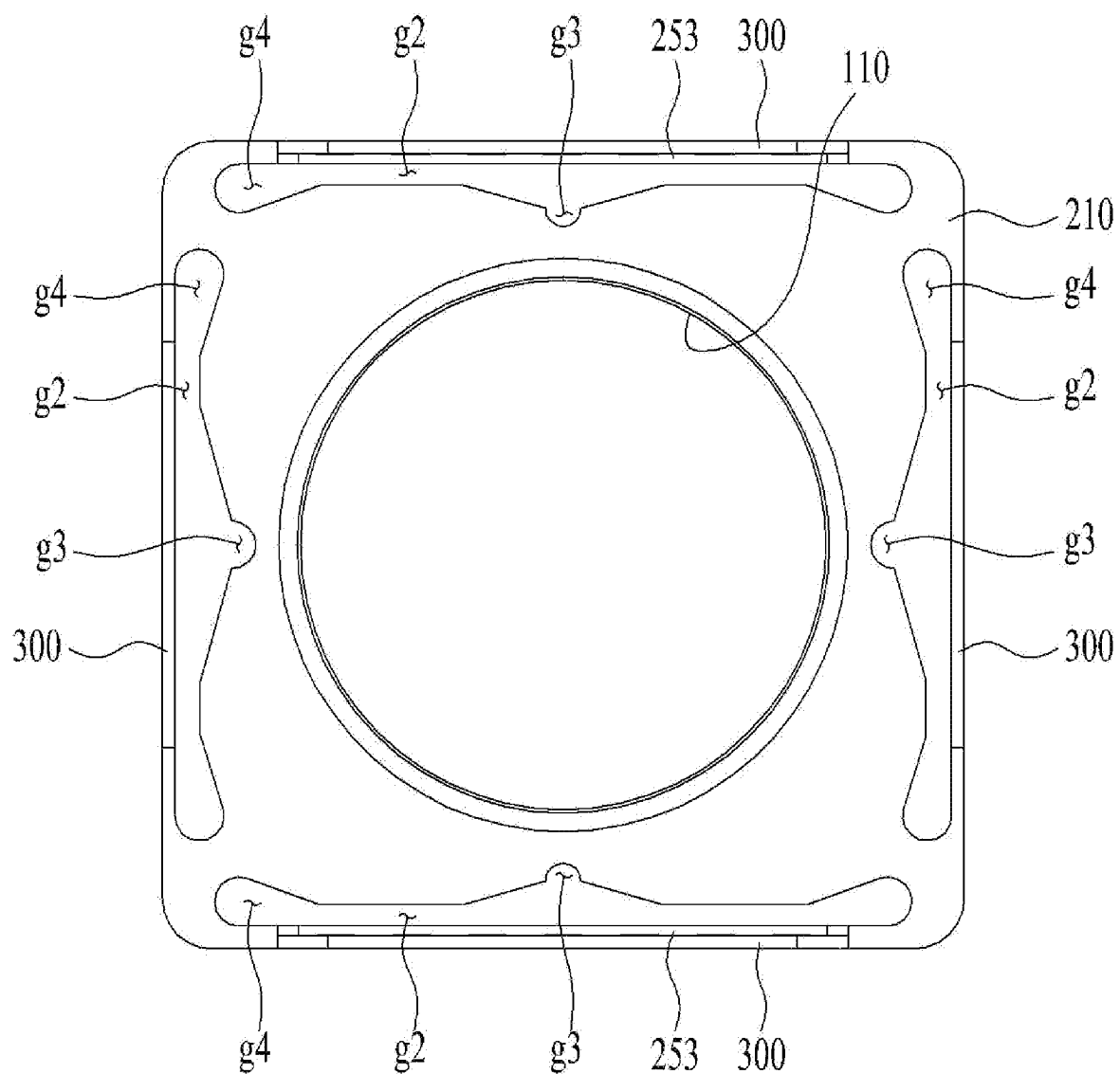


FIG. 28

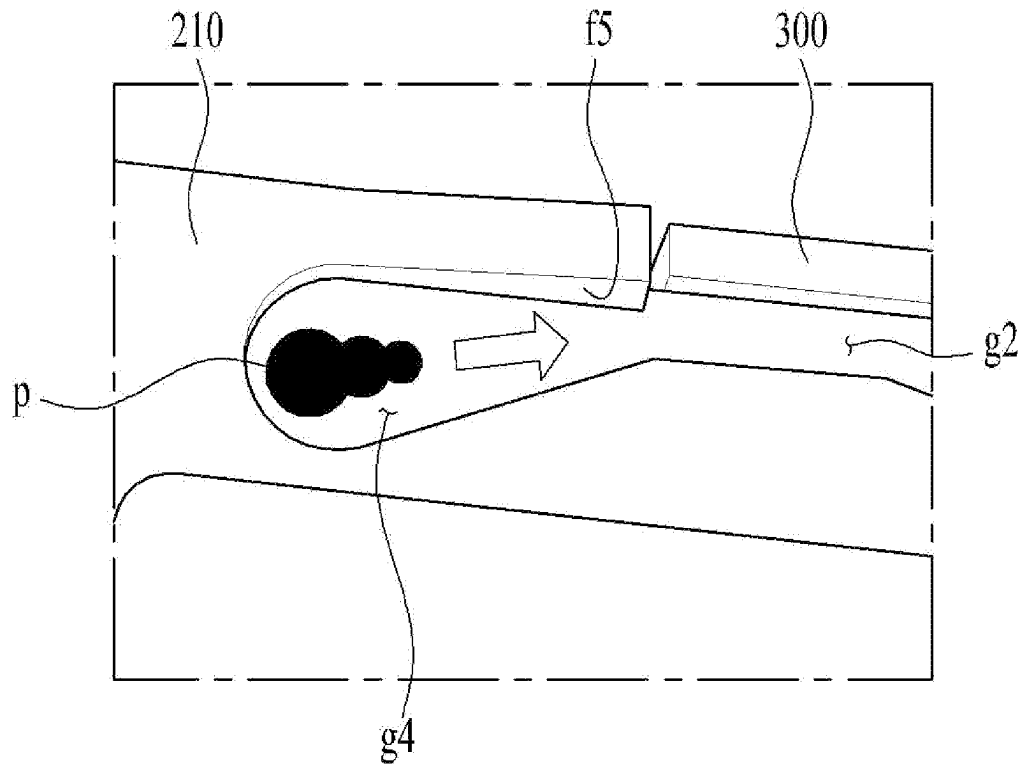
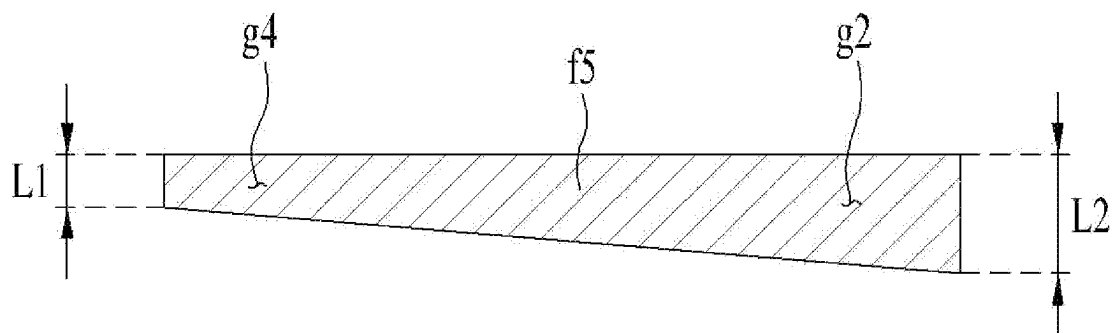


FIG. 29



LENS MOVING APPARATUS

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 16/566,190, filed Sep. 10, 2019; which is a continuation of U.S. patent application Ser. No. 14/884,412, filed Oct. 15, 2015, now U.S. Pat. No. 10,451,891, issued Oct. 22, 2019; which claims priority under 35 U.S.C. § 119 to Korean Patent Application Nos. 10-2014-0140848, filed Oct. 17, 2014, and 10-2014-0158687, filed Nov. 14, 2014, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

Embodiments relate to a lens moving apparatus incorporated in a camera module.

BACKGROUND

Recently, information technology products such as cellular phones, smart phones, tablet PCs and notebook PCs, in which ultracompact digital cameras are incorporated, are being actively developed.

A camera module, which is incorporated in small-sized electronic products such as smart phones, is manufactured by assembling respective components, which have been separately produced, to constitute the camera module.

In the assembly of the camera module, a lens moving apparatus, which constitutes the camera module, may be provided with a magnet for driving a lens, and the magnet may be coupled to a housing by means of an adhesive or the like.

In order to couple the magnet to the housing, an adhesive is first applied to side and upper surfaces of a magnet mounting seat. In the course of such application, it is generally difficult to accurately control the quantity of the adhesive that is applied and to uniformly apply the adhesive to the magnet mounting seat. This is because the viscosity of the adhesive may vary depending on the kind of adhesive, the application temperature and the like.

Accordingly, if the quantity of adhesive that is applied is small or the viscosity of the adhesive is not in the proper range, the adhesive does not uniformly spread, thus causing defective adhesion. In this case, when a drop experiment is performed in order to establish the reliability of the product, defective products in which the magnet is separated from the housing, may be evident.

Once the magnet is bonded to the housing, there is no solution for accurately determining whether a sufficient amount of adhesive has been uniformly applied. Accordingly, there is a need to additionally inject a sufficient amount of adhesive to the bonding area after the magnet is first attached to the housing so as to increase the force of adhesion between the magnet and the housing.

In the operation of attaching the magnet to the housing, when an excessive amount of adhesive is applied to the magnet mounting seat or when the viscosity of the adhesive decreases and the fluidity of the adhesive thus increases, due to the kind of adhesive, the application temperature or the like, even when an appropriate amount of adhesive is applied, the applied adhesive may flow down from the magnet mounting seat.

The adhesive, which has escaped from the magnet mounting seat and cured at another area of the lens moving apparatus, may obstruct movement of other components

constituting the lens moving apparatus, thus causing malfunction or breakage of the lens moving apparatus. A protruding portion, which has been formed differently from intended design due to the cured adhesive, may incur severe deflections in the autofocus function and handshake correction.

When an adhesive flows down and is cured during the procedure of attaching the magnet, the cured adhesive may be removed using tweezers, a knife or the like. However, there is another problem in that other contaminants may adhere to the lens moving apparatus during the removal procedure.

When contaminants infiltrate the lens moving apparatus constituting the camera module in the assembly of the camera module, the infiltrated contaminants may cause malfunction of the lens moving apparatus, quality degradation of captured images or the like. Furthermore, the lens moving apparatus and the camera module have to be firmly assembled.

For these reasons, a bonding operation of filling gaps between respective components with an adhesive may be performed in order to prevent contaminants from infiltrating the gaps while the respective components may be firmly coupled to each other using the adhesive.

The bonding operation has to solve the above problems and has to be rapidly and efficiently performed.

BRIEF SUMMARY

Embodiments intend to provide a lens moving apparatus capable of increasing the bonding force between a magnet and a housing and a camera module including the same.

Furthermore, embodiments intend to provide a lens moving apparatus having a structure capable of firmly coupling respective components to each other and rapidly and efficiently performing a bonding operation of sealing gaps between components and a camera module including the same.

In one embodiment, a lens moving apparatus includes a bobbin on which a first coil is disposed, a first magnet disposed around the bobbin to face the first coil, and a housing, which is disposed to surround at least a portion of the bobbin and has a first magnet mounting seat, which receives the first magnet, wherein the housing is provided with an adhesive inlet, which allows a side portion of the first magnet mounting seat to communicate with an outside surface of the housing.

In another embodiment, a lens moving apparatus includes a housing for supporting a first magnet, a cover member surrounding the housing, a base disposed so as to be spaced apart from the housing by a predetermined distance, a bobbin on which a first coil is provided so as to be moved in a first direction, a support member disposed on a side surface of the housing so as to support the bobbin and the housing while allowing the bobbin and the housing to be movable in second and/or third directions perpendicular to the first direction, and a second coil disposed under the housing to move the support member in the second and/or third directions, wherein the housing includes a first magnet mounting seat for receiving the first magnet and an adhesive inlet allowing the outer surface of the housing to communicate with a side portion of the first magnet mounting seat.

In a further embodiment, a lens moving apparatus includes a printed circuit board, a cover member accommodating at least a portion of the printed circuit board, and a base, on which the printed circuit board is mounted and which is coupled to the cover member, wherein a first recess

is defined by portions of end surfaces of the printed circuit board, the cover member and the base.

BRIEF DESCRIPTION OF THE DRAWINGS

Arrangements and embodiments may be described in detail with reference to the following drawings, in which like reference numerals refer to like elements and wherein:

FIG. 1 is a schematic perspective view showing a lens moving apparatus according to an embodiment;

FIG. 2 is an exploded perspective view showing the lens moving apparatus according to the embodiment;

FIG. 3 is a plan view showing an upper elastic member according to the embodiment;

FIG. 4 is a plan view showing a lower elastic member according to the embodiment;

FIG. 5 is a view showing the lens moving apparatus shown in FIG. 1, from which a cover member has been removed;

FIG. 6 is a perspective view of the bobbin according to one embodiment;

FIG. 7 is a front view showing a support member according to the embodiment;

FIG. 8 is a front view showing a support member according to another embodiment;

FIG. 9 is an exploded perspective view showing a base, a printed circuit board and a second coil according to an embodiment;

FIG. 10 is a perspective view showing a housing according to an embodiment;

FIG. 11 is a bottom perspective view showing the housing according to the embodiment;

FIG. 12 is a view of the housing in which the z-axis direction shown in FIG. 11 is rotated by 180°;

FIGS. 13 and 14 are enlarged views of circle A of FIG. 12;

FIGS. 15 and 16 are views illustrating a second flow restrictor according to one embodiment;

FIG. 17 is a view showing an adhesive inlet according to one embodiment;

FIG. 18 is a view showing an adhesive inlet according to another embodiment;

FIG. 19 is a view illustrating a second flow restrictor according to another embodiment;

FIG. 20 is a schematic perspective view showing the lens moving apparatus according to another embodiment;

FIG. 21 is an enlarged view showing area A of FIG. 20;

FIG. 22 is a view showing a bonding procedure of coupling the lens moving apparatus according to another embodiment;

FIG. 23 is a bottom view showing a base according to one embodiment;

FIG. 24 is a bottom view showing a base according to another embodiment;

FIG. 25 is a bottom view showing a base according to a further embodiment;

FIG. 26 is a bottom view showing a base according to still another embodiment;

FIG. 27 is a bottom view showing a base according to yet another embodiment;

FIG. 28 is a view showing a second recess; and

FIG. 29 is a schematic view showing a side surface of the second recess.

DETAILED DESCRIPTION

Hereinafter, embodiments will be described with reference to the attached drawings. In the drawings, the same or

similar elements are denoted by the same reference numerals even though they are depicted in different drawings. In the following description, a detailed description of known functions and configurations incorporated herein will be omitted when it may make the subject matter of the disclosure rather unclear. Those skilled in the art will appreciate that some features in the drawings are exaggerated, reduced, or simplified for ease in description, and drawings and elements thereof are not shown always at the proper scale.

For reference, in the respective drawings, a rectangular coordinate system (x, y, z) may be used. In the drawings, the x-axis and the y-axis indicate a plane perpendicular to the optical axis, and for convenience, the optical axis (z-axis) direction may be referred to as a first direction, the x-axis direction may be referred to as a second direction, and the y-axis direction may be referred to as a third direction.

FIG. 1 is a schematic perspective view showing a lens moving apparatus according to an embodiment. FIG. 2 is an exploded perspective view showing the lens moving apparatus according to the embodiment. FIG. 3 is a plan view showing an upper elastic member 150 according to the embodiment. FIG. 4 is a plan view showing a lower elastic member 160 according to the embodiment. FIG. 5 is a view showing the lens moving apparatus shown in FIG. 1, from which a cover member 300 has been removed.

A handshake correction apparatus, which is applied to compact camera modules of mobile devices such as smart phones or tablet PCs, refers to an apparatus configured to prevent the contour of an image captured when taking a still picture from not being clearly formed due to vibration caused by user handshake. In addition, an autofocus apparatus is configured to automatically focus the subject image on the surface of an image sensor (not shown). The handshake correction apparatus and the autofocus apparatus may be configured in various manners. In this embodiment, the handshake correction and/or autofocus operations may be performed in such a manner as to move an optical module composed of a plurality of lenses in a first direction or in a plane perpendicular to a first direction.

As shown in FIGS. 1 and 2, the lens moving apparatus according to the embodiment may include a movable unit 100. The movable unit 100 may fulfill the functions of autofocus and handshake correction of a lens.

As shown in FIG. 2, the movable unit 100 may include a bobbin 110, a first coil 120, a first magnet 130, a housing 140, an upper elastic member 150 and a lower elastic member 160.

The bobbin 110 may be provided on the outer surface thereof with the first coil 120 disposed in the first magnet 130, and thus the bobbin 110 may be installed in a space inside the housing 140 so as to be reciprocated in a first direction by the electromagnetic interaction between the first magnet 130 and the first coil 120. Since the bobbin 110 is provided at the outer surface thereof with the first coil 120, the electromagnetic interaction may occur between the first coil 120 and the first magnet 130.

The bobbin 110 may be elastically supported by upper and lower elastic members 150 and 160 such that the autofocus function is fulfilled by the movement of the bobbin 110 in the first direction.

The bobbin 110 may include a lens barrel (not shown) including at least one lens installed therein. The lens barrel may be installed in the bobbin 110 in various manners.

For example, the lens barrel may be coupled to the bobbin 110 in such a manner that a female threaded portion is formed on an inner surface of the bobbin 110 and a male threaded portion, corresponding to the female threaded

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portion, is formed on the outer surface of the lens barrel to engage therewith. However, the disclosure is not limited thereto, and the lens barrel may be directly installed in the bobbin 110 by ways other than the threaded engagement by formation of the threaded portion on the inner surface of the bobbin 110. Alternatively, at least one lens may also be integrally formed with the bobbin 110 without the lens barrel.

The lens coupled to the lens barrel may be composed of a single lens, or may be composed of two or more lenses constituting an optical system.

An autofocus function may be controlled by the direction of electric current, and may be fulfilled by moving the bobbin 110 in the first direction.

For example, the bobbin 110 may move upward from its initial position upon the application of forward current, whereas the bobbin 110 may move downward from its initial position upon the application of reverse current. In addition, the moving distance of the bobbin 110 from its initial position in one direction may be increased or decreased by controlling the amount of current flowing in the one direction.

The bobbin 110 may be provided at upper and lower surfaces thereof with an upper support protrusion 113 (see FIG. 5) and a lower support protrusion (not shown). The upper support protrusion 113 may be configured to have a cylindrical shape or a polygonal column shape such that the inner frame 151 of the upper elastic member 150 is coupled and secured to the bobbin 110.

According to this embodiment, the inner frame 151 may have a first through hole 151a formed at a position corresponding to the upper support protrusion 113.

The upper support protrusion 113 may be inserted in the first through hole 151a and may be secured thereto by means of thermal fusion or an adhesive material such as epoxy. The upper support protrusion 113 may include a plurality of upper support protrusions. The distance between the respective upper support protrusions 113 may be appropriately determined within the range within which interference with peripheral components can be avoided.

Specifically, the upper support protrusions 113 may be symmetrically arranged at a constant interval around the center of the bobbin 110, or may be arranged at irregular intervals so as to be symmetrical with respect to a specific imaginary line extending through the center of the bobbin 110.

The lower support protrusion may be configured to have a cylindrical shape or a polygonal column shape, similarly to the upper support protrusion 113, such that the inner frame 161 of the lower elastic member 160 is coupled and secured to the bobbin 110.

According to the embodiment, the inner frame 161 may have a third through hole 161a formed at a position corresponding to the lower support protrusion. The lower support protrusion may be inserted into the third through hole 161a and may be secured thereto by means of thermal fusion or an adhesive material such as epoxy.

The distance between the respective lower support protrusions may be appropriately determined within the range within which interference with peripheral components can be avoided. In other words, the lower support protrusions may be symmetrically arranged at a constant interval around the center of the bobbin 110.

As shown in FIGS. 3 and 4, the upper elastic member 150 and the lower elastic member 160 may elastically support the upward and/or downward movement of the bobbin 110

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in the first direction. The upper elastic member 150 and the lower elastic member 160 may be constituted by leaf springs.

The upper elastic member 150, which is disposed over the bobbin 110, may be configured such that the inner frame 151 is coupled to the bobbin 110 and the outer frame 152 is coupled to the housing 140. The lower elastic member 160, which is disposed under the bobbin 110, may be configured such that the inner frame 161 is coupled to the bobbin 110 and the outer frame 162 is coupled to the housing 140.

The upper elastic member 150 and the lower elastic member 160s may include the inner frames 151 and 161 coupled to the bobbin 110, the outer frames 152 and 162 coupled to the housing 140, and the connecting members 153 and 163 connected between the inner frames 151 and 161 and the outer frames 152 and 162, respectively.

The connecting members 153 and 163 may be bent at least once to form a predetermined pattern. By virtue of positional change and fine deformation of the connecting members 153 and 163, the upward and/or downward movement of the bobbin 110 in the first direction may be elastically supported.

According to the embodiment, the upper elastic member 150 has a plurality of second through holes 152a formed in the outer frame 152 and a plurality of first through holes 151a formed in the inner frame 151, as shown in FIG. 3.

The second through holes 152a may be fitted over the upper frame support protrusions 144 (see FIG. 10) provided on the upper surface of the housing 140, and the first through holes 151a or recesses may be fitted over the upper support protrusions 113 (see FIG. 6) provided on the upper surface of the bobbin 110. In other words, the outer frame 152 may be fixedly coupled to the housing 140 through the second through holes 152a, and the inner frame 151 may be fixedly coupled to the bobbin 110 through the first through holes 151a or recesses.

The connecting member 153 may connect the inner frame 151 to the outer frame 152 such that the inner frame 151 is elastically deformable with respect to the outer frame 152 within a predetermined range in the first direction.

At least one of the inner frame 151 and the outer frame 152 of the upper elastic member 150 may be provided with at least one terminal conductively connected to at least one of the first coil 120 of the bobbin 110 and the printed circuit board 250.

As shown in FIG. 4, the lower elastic member 160 has a plurality of fourth through holes 162a or recesses formed in the outer frame 162 and a plurality of third through holes 161a or recesses formed in the inner frame 161. The lower elastic member 160 can include a first upper elastic member 160a and a second upper elastic member 160b, which are separated from each other.

The fourth through holes 162a or recesses may be fitted over the lower frame support protrusions 145, and the third through holes 161a or recesses may be fitted over the lower support protrusions provided on the lower surface of the bobbin 110.

In other words, the outer frame 162 may be fixedly coupled to the housing 140 through the fourth through holes 162a or recesses, and the inner frame 161 may be fixedly coupled to the bobbin 110 through the third through holes 161a or recesses.

The connecting members 163 may connect the inner frame 161 to the outer frame 162 such that the inner frame 161 is elastically deformable with respect to the outer frame 162 in the first direction within a predetermined range.

As shown in FIG. 3, the upper elastic member **150** may include a first upper elastic member **150a** and a second upper elastic member **150b**, which are separated from each other. Thanks to the dual partitioning structure, current having different polarities or different electric powers may be applied to the first upper elastic member **150a** and the second upper elastic member **150b** of the upper elastic member **150**.

Specifically, the inner frame **151** and the outer frame **152** are coupled to the bobbin **110** and the housing **140**, respectively, and then solder portions are provided at positions corresponding to opposite ends of the first coil **120** disposed at the bobbin **110**. Subsequently, conductive connections such as solder are provided at the solder portions, whereby current having different polarities or different electric powers may be applied to the first upper elastic member **150a** and the second upper elastic member **150b**.

In addition, the first upper elastic member **150a** is conductively connected to one of opposite ends of the first coil **120**, and the second upper elastic member **150b** is conductively connected to the second upper elastic member **150b**, thus enabling external current and/or voltage to be applied thereto.

The upper elastic member **150** and the lower elastic member **160** may be assembled to the bobbin **110** and the housing **140** through a bonding process by means of thermal fusion and/or adhesive. In some cases, the assembly operation comprises thermal fusion and then bonding by adhesive, performed in that order.

In a modification thereof, the lower elastic member **160** may be configured to have a dual partitioning structure, and the upper elastic member **150** may be configured as an integral structure.

At least one of the inner frame **161** and the outer frame **162** of the lower elastic member **160** may be conductively connected to at least one of the first coil **120** of the bobbin **110** and the printed circuit board **250**. The printed circuit board **250** may include the at least one terminal member **253**, which may be conductively connected to the elastic members **150** and **160**.

The printed circuit board **250** may be coupled to the upper surface of the base **210**. As shown in FIGS. 2 and 5, the printed circuit board **250** may have a through hole through which a support member mounting recess **214** is exposed.

The printed circuit board **250** may be provided with a surface on which the bent terminal member **253** is mounted. A plurality of terminals may be disposed on the surface on which the first terminal member **253** is mounted so as to supply electric power, supplied from the outside, to the first coil **120**. The number of the terminals formed on the terminal member **253** may be increased or decreased in accordance with the components that are required to be controlled. The printed circuit board **250** may further include an additional bent surface and an additional terminal member.

The base **210** is disposed under the bobbin **110**. As shown in FIGS. 2 and 5, the base **210** may be configured to have an approximately rectangular shape, and may be provided on a flat surface thereof with the support member **220** held thereto. The base **210** may be provided with stepped portions **211** to which an adhesive is applied in order to adhesively attach the cover member **300** thereto. The surface of the stepped portion **211** may contact one end of the cover member **300**.

The base **210** may have a support recess, which is formed in the surface that faces the terminal member **253** of the printed circuit board **250** and has a size corresponding to the

first terminal member **251**. The support recess may be recessed from the outer surface of the base **210** by a predetermined depth such that the portion of the terminal member **253** that protrudes from the base **210** is eliminated or controlled to a desired amount.

The stepped portions **211** may guide the cover member **300**, which is mounted on the base **210**. The cover member **300** may be mounted on the base **210** such that an end of the cover member **300** engages with the base **210** in a surface-contact manner. The stepped portions **211** and the end of the cover member **300** may be adhesively attached to each other or sealed by means of, for example, an adhesive.

The base **210** may be provided at corners of an upper surface thereof with the support member mounting recess **214**, in which the support member **220** is inserted. The support member mounting recess **214** may be provided with adhesive so as to securely hold the support member **220**.

The end of the support member **220** may be inserted or disposed in the support member mounting recess **214**, and may then be secured thereto by means of an adhesive or the like. The support member mounting recess **214** may include one or more support member mounting recesses formed in the flat surface at which the support member **220** is installed. The support member mounting recess **214** may have an approximately rectangular shape.

As shown in FIGS. 2 and 5, according to the embodiment, the support member mounting recesses **214** may be provided in the base **210** in such a manner that two support member mounting recesses **214** are provided in each flat surface. The number of support member mounting recesses **214** may be increased or decreased in accordance with the shape of the support member **220**, and three or more support member mounting recesses **214** may be provided in each flat surface.

The cover member **300** may be configured to have an approximate box shape capable of accommodating the movable unit **100**, the printed circuit board **250** and the base **210**. As shown in FIG. 1 etc., the cover member **300** may have escaping portions or recesses formed in positions corresponding to the stepped portions **211** of the base **210**, and, as such, adhesive and the like may be injected through the escaping portions or recesses.

At this point, the adhesive is set to have a low viscosity such that the adhesive injected through the escaping portions or recesses can infiltrate the contact areas between the stepped portions **211** and the end of the cover member **300**. The adhesive applied to the escaping portions or recesses fills the gap between the mating surfaces of the cover member **300** and the base **210** through the escaping portions of recesses, thus enabling the cover member **300** to be sealingly coupled to the base **210**.

FIG. 6 is a perspective view of the bobbin **110** according to one embodiment. The bobbin **110** may include a first stopper **111** and/or a second stopper **112**.

The first stopper **111** may prevent the upper surface of the bobbin **110** from directly colliding with the inner surface of the cover member **300** shown in FIG. 1 even when the bobbin **110** moves beyond the specified range due to external impact or the like in the course of moving in the first direction for the autofocus function. Furthermore, the first stopper **111** may also serve to guide the installation position of the upper elastic member **150**.

According to the embodiment, the first stopper **111** may include a plurality of stoppers which protrude upward by a first height h_1 , as shown in FIG. 6. Specifically, at least four first stoppers having a polygonal column shape may protrude. The first stoppers **111** may be configured to be

symmetrical about the center of the bobbin 110, or may be configured to be asymmetrical, as shown in the drawing.

The second stopper 112 may prevent the lower surface of the bobbin 110 from directly colliding with the upper surfaces of the base 210 and the circuit board 250 shown in FIG. 2 even when the bobbin 110 moves beyond the specified range due to external impact or the like in the course of moving in the first direction for the autofocus function.

According to the embodiment, the second stopper 112 may circumferentially protrude from the side surface of the upper portion of the corner of the bobbin 110, and the housing 140 may have a bobbin mounting seat 146 (see FIG. 10) formed at a position corresponding to the second stopper 112.

When the state in which the stopper 112 and the lower surface 146a of the bobbin mounting seat 146 (see FIG. 10) are in contact with each other is set to be the initial position, the autofocus function may be realized by unidirectional control in which the bobbin 110 is raised when current is supplied to the first coil 120 and is lowered when the supply of the current is halted.

If the state in which the stopper 112 is spaced apart from the lower surface 146a of the bobbin mounting seat 146 is set to be the initial position, the autofocus function may also be realized by bidirectional control in which the bobbin 110 is moved upward or downward in the first direction depending on the direction of current. For example, the bobbin 110 may be moved upward upon the application of forward current and may be moved downward upon the application of reverse current.

The bobbin mounting seat 146 of the housing 140, which corresponds to the second stopper 112, is recessed. As shown in FIG. 10, the width w2 of the bobbin mounting seat 146 is configured to have a predetermined tolerance with respect to the first width w1 of the second stopper 112 so as to prevent the second stopper 112 from rotating in the bobbin mounting seat 146. Accordingly, the second stopper 112 may prevent rotation of the bobbin 110 even if the bobbin 110 is subjected to force that is applied in the direction of rotating about the z axis, rather than in the first direction.

The bobbin 110 may include two winding protrusions 115 formed on the outer surface of an upper portion thereof. The two ends, that is, the starting line and ending line of the first coil 120, may be wound around respective winding protrusions 115. The two ends of the first coil 120 may be conductively connected to the upper surface of the upper elastic member 150 at a position on the bobbin 110 close to the winding protrusions 115, by means of a conductive connecting member such as solder.

The winding protrusions 115 may include a pair of winding protrusions which are disposed at positions symmetrical about the center of the bobbin 110, or the pair of winding protrusions 115 may be disposed close to each other.

The winding protrusions 115 may be respectively provided at the ends thereof with retaining protrusions 115a so as to prevent the first coil 120 wound therearound from being separated therefrom or to guide the position of the first coil 120. The winding protrusions 115 may have a width that increases radially outward from the outer circumferential surface of the bobbin 110, and may be provided at the ends thereof with the stepped retaining protrusions 115a.

As shown in FIG. 6, the bobbin 110 may be provided with at least one groove 116 close to the winding protrusions 115. The groove 116 may include a first groove 116a and/or a second groove 116b. The first groove 116a and/or the second

groove 116b may have a depth and width which are greater than the diameter of the first coil 120 such that the starting line or the ending line of the first coil 120 passes through the groove.

Accordingly, the starting line and the ending line of the first coil 120 may be easily received in the first and/or second grooves 116a and 116b. Furthermore, the starting line and the ending line of the first coil 120 may extend through the first and/or second grooves 116a and 116b without interference with the upper elastic member, which is disposed thereon.

FIG. 7 is a front view showing the support member 220 according to the embodiment. FIG. 8 is a front view showing a support member 220 according to another embodiment. FIG. 5 illustrates the support member 220 according to one embodiment, which is disposed at the normal position.

The support member 220 is disposed at one side surface of the housing 140. The support member 220 is coupled at an upper portion thereof to the housing 110 and is coupled at a lower portion thereof to the base 210. The support member 220 may support the bobbin 110 and the housing 140 such that the bobbin 110 and the housing 140 can be displaced in the second and third directions, which are perpendicular to the first direction. The support member 220 may be conductively connected to the first coil 120.

The support members 220 may be separately disposed on the second surfaces 142 of the housing 140 so as to support the housing 140 in the state of being spaced apart from the base 210 by a predetermined distance. One end of the support member 220 may be inserted or disposed in the support member mounting recess 214, and may then be coupled thereto using an adhesive material such as epoxy. The other end of the support member 220 may be secured to an upper end of the side wall of the housing 140.

Since the support member 220 is disposed on the second surfaces 142 of the housing 140, a total of four support members may be symmetrically disposed on respective side surfaces 142 one on each. However, the disclosure is not limited thereto, and a total of eight support members 220 may be provided, two on each flat surface. The support member 220 may be conductively connected to the upper elastic member 150 or to the flat surface of the upper elastic member 150.

Specifically, the support member 220 may include a first coupling portion 221, a second coupling portion 224, a first elastic deformation portion 222, a second elastic deformation portion 223 and a connecting portion 225.

The first coupling portion 221 is the portion of the support member 220 that is coupled to the upper end of the second surface 142 of the housing 140. The first coupling portion 221 may include through holes at locations corresponding to the protrusions protruding from the second surface 142 of the housing 140 such that the coupling portion 221 is coupled to the upper end of the housing 140 by fitting the protrusions into the through holes.

Since the support member 220 is constructed separately from the upper elastic member 150, the support member 220 and the upper elastic member 150 may be conductively connected to each other by means of conductive adhesive, solder, welding or the like. Accordingly, the upper elastic member 150 may apply current to the first coil 120 through the support member 220, which is conductively connected thereto.

The second coupling portion 224 may be the portion that is coupled to the base 210, and may be provided at the end of the support member 220. Although the second coupling portion 224 may be configured as a plate having a greater

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width than that of the first and second elastic deformation portions **222** and **223**, the second coupling portion **224** may have a width equal to or smaller than the width of the first and second elastic deformation portions **222** and **223**, without being limited thereto.

According to the embodiment, the second coupling portion **224** may be divided into two elements, and may be inserted or disposed in the support member mounting recess **214**, as shown in FIGS. **8** and **9**. The second coupling portion **224** may be securely coupled to the support member mounting recess **214** of the base **210** by means of an adhesive material such as epoxy.

However, the disclosure is not limited thereto, and the support member mounting recess **214** may be configured to correspond to the second coupling portion **224** and may be fitted in the second coupling portion **224**. The second coupling portion **224** may include a single second coupling portion or two or more second coupling portions. The support member mounting recess **214** may be formed in the base **210** in accordance with the number of second coupling portion **224**.

The elastic deformation portions **222** and **223** may be bent at least once to provide a determined pattern. According to the embodiment, the elastic deformation portions may include first and/or second elastic deformation portions **222**, **223**. The first elastic deformation portion **222** extends from the first coupling portion **221** and may be connected to the connecting portion **225**. The second elastic deformation portions **223** may extend from the second coupling portion **224**, and may be connected to the connecting portion **225**.

The elastic deformation portions **222** and **223** may be positioned with the connecting portion **225** disposed therebetween, and may be configured to have a symmetrical shape. As shown in FIGS. **7** and **8**, when the first elastic deformation portion **222** is configured as two or more bent portions in a zigzag manner, the second elastic deformation portion **223** may also be correspondingly configured. However, the disclosure is not limited thereto, and the first elastic deformation portion **222** may be provided alone, or the second elastic deformation portion **223** may be configured so as to have some other structure.

The above configuration is merely an example, and the embodiment may be configured to have various patterns, such as a zigzag pattern. In this case, only one elastic deformation portion may be provided, without being divided into the first and second elastic deformation portions **222** and **223**, and the elastic deformation portion may be configured to have a suspension wire shape, rather than the above pattern.

According to the embodiment, straight portions of the first and second elastic deformation portions **222** and **223** may be approximately parallel to a plane that is perpendicular to the first direction.

When the housing **140** moves in the second and/or third directions, which are perpendicular to the first direction, the elastic deformation portions **222** and **223** may be elastically and finely deformed in the moving direction of the housing **140** or in the longitudinal direction of the support member **220**.

As a result, since the housing **140** may move in the second and third directions with almost no displacement in the first direction, the accuracy of handshake correction may be improved. This utilizes the property of the first and second elastic deformation portions **222** and **223**, which are capable of extending in a longitudinal direction. The term “longitudinal direction” may refer to the connecting direction between first and second coupling portions **221** and **224**.

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The connecting portion **225** may include a pair of connecting portions, which are disposed to be symmetrical to each other and connect the first and second elastic deformation portions **222** and **223**. Although the connecting portion **225** may be disposed between the first and second elastic deformation portions **222** and **223**, as described above, the disclosure is not limited thereto. The connecting portion **225** may also be connected to one elastic deformation portion.

Although one support member **220** is provided with a pair of first and second elastic deformation portions **222** and **223** in the embodiment, the first and second coupling portions **221** and **224** may be integrally constructed, and the pair of first and second elastic deformation portions **222** and **223** may be coupled to the housing **140** and the base **210** at the same time.

FIG. **9** is an exploded perspective view showing the base **210**, the printed circuit board **250** and a second coil **230** according to an embodiment. The lens moving apparatus may further include the second coil **230** and a detection sensor **240**.

The second coil **230** may perform handshake correction by moving the housing **140** in the second and/or third directions by electromagnetic interaction with the first magnet **130**. Accordingly, there is a need to mount the first magnet **130** at a position corresponding to the second coil **230**.

The second coil **230** may be disposed so as to face the first magnet **130** secured to the housing **140**. By way of example, the second coil **230** may be disposed outside the first magnet **130**, or may be disposed under the first magnet **130**, with a predetermined distance therebetween.

According to the embodiment, although a total of four second coils **230** may be respectively disposed on the four corners of the printed circuit board **250**, the disclosure is not limited thereto. Only one second coil for movement in the second direction and one second coil for movement in the third direction may be disposed, or four or more second coils may be disposed.

In this embodiment, although a circuit pattern having the shape of the second coil **230** may be formed on the printed circuit board **250** and an additional second coil **230** may be disposed over the printed circuit board **250**, the disclosure is not limited thereto. Alternatively, only an additional second coil **230** may be disposed over the printed circuit board **250**, without forming a circuit pattern having the shape of the second coil **230** on the printed circuit board **250**.

Furthermore, the second coil **230**, which has been prepared by winding a wire into a doughnut shape or which has the shape of a finely patterned coil, may be conductively connected to the printed circuit board **250**.

A circuit member **231** including the second coil **230** may be disposed on the upper surface **250a** of the printed circuit board **250** that is positioned over the base **210**. However, the disclosure is not limited thereto, and the second coil **230** may be disposed on the base **210** in a state of being in close contact therewith, or may be spaced apart from the base **210** by a predetermined distance. In other examples, a substrate on which the second coil is formed may be layered on the printed circuit board **250** and connected thereto.

The detection sensor **240** is positioned at the center of the second coil **230** so as to detect the movement of the housing **140**. The detection sensor may fundamentally serve to detect movement of the housing **140** in the first direction, and in some case may serve to detect movement of the housing **140** in the second and/or third directions.

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The detection sensor **240** may be constituted by a Hall sensor, but may be constituted by any sensor as long as the sensor is able to detect variation of magnetic force. As shown in FIG. 9, the detection sensor **240** may include two detection sensors, which are mounted on the corners of the base **210** positioned under the printed circuit board **250**. The mounted detection sensors **240** may be received in sensor mounting recesses **215** formed in the base **210**. The lower surface of the printed circuit board **250** may be the surface opposite the surface on which the second coil **230** is disposed.

The detection sensor **240** may be disposed under the second coil **230** with the printed circuit board **250** disposed therebetween such that the detection sensor **240** is spaced apart from the second coil **230** by a predetermined distance. In other words, the detection sensor **240** is not directly connected to the second coil **230**, and the printed circuit board **250** may be provided on the upper surface thereof with the second coil **230** and on the lower surface thereof with the detection sensor **240**.

FIG. 10 is a perspective view showing the housing **140** according to an embodiment. FIG. 11 is a bottom perspective view showing the housing **140** according to the embodiment.

The housing **140** may be configured to have a hollow column shape supporting the first magnet **130**. According to the embodiment, the housing **140** may have an approximately octagonal shape, as shown in FIGS. 10 and 11. The housing **140** may include a first side surface **141** and a second side surface **142**. The first magnet **130** may be mounted on the first side surface **141**, and the support member **220** may be disposed on the second side surface **142**.

The first side surface **141** may be positioned at the corner of the housing **140**. According to the embodiment, the first side surface **141** may be configured to have a surface area equal to or larger than that of the first magnet **130**. The first magnet **130** may be secured in a first magnet mounting seat **141a** formed in the inner surface of the first side surface **141**.

The first magnet mounting seat **141a** may be constituted by a recess having a size corresponding to the size of the first magnet **130**, and may face at least three surfaces of the first magnet **130**, that is, both lateral surfaces and the upper surface of the first magnet **130**.

Although the first magnet **130** may be secured to the first magnet mounting seat **141a** by means of an adhesive, the disclosure is not limited thereto, and double-sided adhesive tape or the like may be alternatively be used. Furthermore, the first magnet mounting seat **141a** may not be constituted by the recess shown in FIG. 4, but may be configured to have a mounting hole in which the first magnet **130** is partially fitted or exposed.

The housing **140** may be provided on the upper surface thereof with a plurality of third stoppers **143**. The third stoppers **143** may restrict the upward movement of the housing **140**.

The third stoppers **143** may also serve to guide the installation position of the upper elastic member **150**. To this end, the upper elastic member **150** may be provided at positions corresponding to the third stopper **143** with guide holes **155** having a shape corresponding to the third stopper **143**, as shown in FIG. 3.

Although the first side surface **141** may be disposed parallel to the side surface of the cover member **300**, the disclosure is not limited thereto, and the first side surface **141** may be configured to have a larger surface area than the second side surface **142**.

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As shown in FIGS. 10 and 11, the second side surface **142** may be recessed to define a reception recess **142a** having a predetermined depth. According to the embodiment, the lower surface of the reception recess **142** may be open, and both the upper and lower surfaces of the reception recess **142** may be open.

Thanks to the provision of the reception recess **142a**, spatial interference between the connecting members **153** and **163** and the bobbin **110** is eliminated when the bobbin **110** moves with respect to the housing **140** in the first direction, thus allowing the connecting members **153** and **163** to be easily elastically deformed.

The lower open end of the reception recess **142a** may prevent the second coupling portion **224** of the support member **220** from interfering with the housing **140**. As shown in FIG. 11, the reception recess **142a** may be provided at the upper end thereof with a stepped portion **142b** so as to partially support the upper portion of the support member **220**.

Although the reception recess **142a** may be positioned at the lateral side of the housing **140** as in the embodiment, the reception recess **142a** may alternatively be positioned at the corner of the housing **140** depending on the shape and/or position of the connecting members **153** and **163** of the elastic members **150** and **160** and the support member **220**.

The housing **140** may be provided at the upper surface thereof with a plurality of upper frame support protrusions **144** to which the outer frame **152** of the upper elastic member **150** is coupled. The number of upper frame support protrusions **144** may be greater than the number of upper support protrusions **113**. This is because the length of the outer frame **152** is longer than the length of the inner frame **151**.

The outer frame **152** may have second through holes **152a** which are formed at positions corresponding to the upper frame support protrusions **144**, and may have a shape corresponding to the upper support protrusions **144**. The upper frame support protrusions **144** may be fixed in the second through holes **152a** by means of an adhesive or thermal fusion.

As shown in FIG. 11, the housing **140** may be provided at a lower surface thereof with a plurality of lower frame support protrusions **145** to which the outer frame **162** of the lower elastic member **160** is coupled. The lower frame support protrusions **145** may be more numerous than the lower support protrusions because the outer frame **162** of the lower elastic member **160** is longer than the inner frame **161**.

The outer frame **162** may have fourth through holes **162a**, which are formed at positions corresponding to the lower frame support protrusions **145** and have a shape corresponding to that of the lower frame support protrusions **145**. The lower frame support protrusions **145** may be fixed in the fourth through holes **162a** by means of an adhesive or thermal fusion.

The housing **140** may further be provided at a lower surface thereof with fourth stoppers **147**. The fourth stoppers **147** may serve to restrict the downward moving distance of the housing **140**. Therefore, the fourth stoppers **147** may prevent the bottom surface of the housing **140** from colliding with a base **210** and/or a printed circuit board **250**.

The fourth stoppers **147** may be maintained in the state of being spaced a predetermined distance apart from the base **210** and/or the printed circuit board **250** during an initial stage or during normal operation. By virtue of this construction, the housing **140** may be spaced apart not only from the base **210** disposed thereunder but also from the cover member **300** disposed thereover, and, as such, the housing

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140 may be maintained at a constant level in the first direction without interference from obstacles above and below. Accordingly, the housing 140 may perform a shifting action in second and third directions, perpendicular to the first direction.

FIG. 12 is a view of the housing in which the z-axis direction shown in FIG. 11 is rotated by 180°. When the first magnet 130 is coupled to the first magnet mounting seat 141a using an adhesive g, the housing 140 may be installed downward, unlike the disposition shown in FIGS. 1 and 2, and the coupling operation may be performed. Since the object of the embodiment is to solve the problems occurring in the case where an insufficient or excessive amount of adhesive g is applied to the first magnet mounting seat 141a, in order to clearly explain the related structures, figures in which the first direction is directed downward are illustrated in FIG. 12 and subsequent drawings.

As shown in FIG. 12, the housing 140 may have adhesive inlets which are formed between the outer surface of the housing 140 and the side surface P1 of the first magnet mounting seat 141a. In other words, the housing 140 may have adhesive inlets that are depressed from the lower surface of the housing 140 so as to allow the first magnet mounting seat 141a to communicate with the outside of the housing 140.

As shown in FIG. 12, the adhesive inlet 141b may be configured to have a recess which is formed by partially depressing the lower surface of the housing 140. However, this is merely one embodiment, and the adhesive inlet 141b may also be configured to have a hole shape which connects the outer surface of the housing 140 and the side surface P1 of the first magnet mounting seat 141a. In this case, the hole shape may be open at a portion of the inner surface thereof.

The adhesive inlet 141b is preferably formed at a lower portion of the side surface P1 of the first magnet mounting seat 141a. Specifically, this is because, since the housing 140 is in an inverted state when the first magnet 130 is mounted on the first magnet mounting seat 141a, unlike the disposition shown in FIGS. 1 and 2, the adhesive g, which is introduced through the adhesive inlet 141b, flows toward the upper end from the lower end of the first magnet mounting seat 141a due to gravity.

Specifically, the first magnet 130 may be configured to have, for example, a trapezoidal shape when viewed in a plan view, and the first magnet mounting seat 141a, which is provided at the corner of the housing 140, may also be configured to have a trapezoidal shape when viewed in a plan view. Meanwhile, the side surface P1 of the first magnet mounting seat 141a may include a first side surface P1-1, which corresponds to the upper side of the trapezoid, and second side surfaces P1-2 connected to both side edges of the first side surface P1-1.

The adhesive inlet 141b may be formed between the first side surface P1-1 and/or the second side surface P1-2. The specific example of the magnetic inlet 141b will be described later.

In the procedure of mounting the first magnet 130 on the housing 140, the adhesive g is first applied to the side surface P1 and the upper surface P2 of the first magnet mounting seat 141a. At this time, it is difficult to uniformly apply an accurate quantity of adhesive g to the first magnet mounting seat 141a. This is because the viscosity of the adhesive g may vary depending on the kind of adhesive g, the temperature during the mounting operation, or the like.

Accordingly, when the quantity of adhesive g that is applied is small, the adhesive g may have an unsuitable

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viscosity and thus not uniformly spread, thus causing defective attachment of the first magnet 130. When a drop experiment for determining the reliability of products is performed under such a condition, defective products in which the first magnet 130 is separated from the housing 140 may be evident.

There is no way to determine whether a sufficient amount of adhesive g is uniformly applied once the first magnet 130 is attached to the housing 140. Accordingly, after the first magnet 130 is first attached to the housing 140, there is the need to inject additional adhesive g to the attaching area of the housing 140 in order to supply a sufficient amount of adhesive g to the attaching area, thus increasing the adhesive force between the first magnet 130 and the housing 140.

Accordingly, in order to supply the additional adhesive g to the attaching area after attaching the first magnet 130 to the housing 140, the adhesive inlet 141b may be formed.

At this point, when the adhesive g is injected through the adhesive inlet 141b using an injector n, the adhesive g discharged from the needle point of the injector may penetrate into the gap between the first magnet 130 and the first magnet mounting seat 141a, at which the adhesive g was not applied in the first application, thus increasing the bonding force between the first magnet 130 and the housing 140.

In the embodiment, the adhesive inlet 141b is formed so as to connect the outer surface of the housing 140 and the side surface P1 of the first magnet mounting seat 141a, and the adhesive g is additionally applied to the bonding area through the magnet inlet 141b so as to increase the bonding force between the first magnet 130 and the housing 140.

The upper surface P2 of the first magnet mounting seat 141a may be provided with at least one first flow restrictor 141a-1 for restricting the flow of the adhesive g. The first flow restrictor 141a-1 may be configured to have a groove parallel to the lower side of the trapezoid of the first magnet mounting seat 141a.

When an excessive amount of adhesive g is applied to the first magnet mounting seat 141a or when the viscosity of the adhesive g is decreased and its fluidity is thus increased due to the kind of the adhesive g, the temperature during the bonding procedure, or the like, even if a suitable amount of adhesive g is applied, the applied adhesive g may escape from the first magnet mounting seat 141a and flow down.

Since the adhesive g, which has escaped from the first magnet mounting seat 141a and has cured in another area of the lens moving apparatus, obstructs the movement of other components, it may cause operational malfunction or breakage of the lens moving apparatus. The portions that protrude differently from the original design due to the cured adhesive g may cause severe deflections in the autofocus or hand-shake correction of the lens moving apparatus.

In order to prevent the adhesive g from flowing down, the upper surface P2 of the first magnet mounting seat 141a may be provided with the first flow restrictor 141a-1. A second flow restrictor 146a-1 may further be provided.

FIGS. 13 and 14 are enlarged views of circle A of FIG. 12. Although FIG. 13 illustrates the second flow restrictor 146a-1, the first flow restrictor 141a-1 is described with reference to FIGS. 13 and 14, and the second flow restrictor 146a-1 is described later with reference to FIGS. 15 and 16. In the embodiment, the first flow restrictor 141a-1 and the second flow restrictor 146a-1 may be selectively provided. In other words, one or both of the first flow restrictor 141a-1 and the second flow restrictor 146a-1 may be provided.

As shown in FIGS. 13 and 14, the first flow restrictor 141a-1 may be provided on the upper surface P2 of the first magnet mounting seat 141a at an inner position close to the

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open plane of the first magnet mounting seat **141a** so as to prevent the adhesive **g** applied to the upper surface **P2** or the side surface **P1** of the first magnet mounting seat **141a**, from flowing out of the first magnet mounting seat **141a**.

Specifically, the first flow restrictor **141a-1** may be provided as a groove positioned parallel to the lower side of the trapezoid of the first magnet mounting seat **141a**.

In the cross-sectional view shown in FIG. **13**, although the first flow restrictor **141a-1** is configured to have a semicircular shape or a “U” shape, it may be configured to have various shapes without limitation as long as it can receive the flowable adhesive **g** therein.

As shown in FIG. **13**, by way of example, although the first flow restrictor **141a-1** is constructed to have a pair of parallel grooves, the disclosure is not limited thereto. In other words, the number of first flow restrictors **141a-1** may be one or three or more.

The number of first flow restrictor **141a-1** may be appropriately determined in accordance with the area of the upper surface **P2** of the first magnet seat **141a**, the amount of adhesive **g** that is applied, or the like. Specifically, as the surface area of the upper surface **P2** of the first magnet mounting seat **141a**, the amount of adhesive **g**, or the like is increased, the number of first flow restrictors **141a-1** may be increased.

As shown in the cross-sectional view of FIG. **13(b)**, the upper surface **P2** of the first magnet mounting seat **141a** may be provided with a plurality of stepped portions **s** in order to restrict the flow of the adhesive **g**. Here, since the plurality of stepped portions **s** increases the length of the path through which the adhesive **g** flows on the upper surface **P2**, the plurality of stepped portions **s** may serve to prevent the adhesive **g** from flowing out of the first magnetic mounting seat **141a** and to maintain the adhesive **g** on the first magnet mounting seat **141a**.

Although two stepped portions **s** are, by way of example, illustrated in FIG. **13**, the disclosure is not limited thereto. Specifically, the number of stepped portions **s** may be one or three or more. The stepped portion **s** may be formed on the upper surface **P2**, together with or without the first flow restrictor **141a-1**. In this case, in order to further increase the length of the flowing path of the adhesive **g**, the edge of the stepped portion **s** may be chamfered so as to form an inclined surface similar to the second flow restrictor **146a-1**.

As shown in a cross-sectional view in FIG. **14(a)**, the adhesive **g**, which flows on the side surface **P1** or the upper surface **P2** of the first magnet mounting seat **141a**, flows toward the open plane of the first magnet mounting seat **141a**. The flowing adhesive **g** is first introduced in the first flow restrictor **141a-1**, which is positioned farther away from the open plane of the first magnet mounting seat **141a**, whereby the adhesive **g** cannot advance any further and cannot overflow out of the first magnet mounting seat **141a**.

As shown in a cross-sectional view in FIG. **14(b)**, when too much adhesive **g** is applied or the viscosity of the adhesive **g** is excessively low, the adhesive **g** may further advance toward the open plane of the first magnet mounting seat **141a**, beyond the first flow restrictor **141a-1**.

At this time, the adhesive **g** is introduced into the first flow restrictor **141a-1**, positioned closer to the open plane of the first magnet mounting seat **141a**, whereby the adhesive **g** cannot advance any further and cannot overflow out of the first magnet mounting seat **141a**.

As described above, when the number of first flow restrictors **141a-1** is appropriately selected in consideration of the surface area of the upper surface **P2** of the first magnet mounting seat **141a**, the typical amount of the adhesive **g**, or

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the like, it is possible to efficiently prevent the adhesive **g** from overflowing out of the first magnet mounting seat **141a**.

According to the embodiment, thanks to the provision of the first flow restrictor **141a-1** formed in the upper surface **P2** of the first magnet mounting seat **141a**, it is possible to prevent the adhesive **g** from overflowing out of the first magnet mounting seat **141a**.

Accordingly, since it is possible to prevent the adhesive **g** from overflowing out of the first magnet mounting seat **141a** and being cured, malfunctions in autofocus, handshake correction or the like of the lens moving apparatus due to protruding portions attributable to the cured adhesive **g** can be prevented.

FIGS. **15** and **16** are views illustrating the second flow restrictor **146a-1** according to one embodiment. The housing **140** may be provided over the first magnet mounting seat **141a** thereof with a bobbin mounting seat **146** in which the bobbin **110** is partially received, and the area of the housing **140** where the inner surface of the housing and the bobbin mounting seat **146** meet in the first direction may be provided with the second flow restrictor **146a-1** for restricting the flow of the adhesive **g**.

Specifically, the second flow restrictor **146a-1** may be formed at the area where the lower surface **146a** of the bobbin mounting seat **146** meets the inner surface of the housing **140**.

The second flow restrictor **146a-1** may be configured as an inclined surface when viewed in the first direction. By way of example, the second flow restrictor **146a-1** may be formed into the inclined surface through chamfering.

The second stopper **112**, which is a portion of the bobbin **110**, may be mounted in the bobbin mounting seat **146**. The bobbin mounting seat **146** may serve to restrict the movable range of the second stopper **112**.

The second stopper **112** may not be securely mounted in the bobbin mounting seat **146**, but may be movable within the volume range defined by the bobbin mounting seat **146**. Accordingly, when the adhesive **g** flows into the bobbin mounting seat **146** from the first magnet mounting seat **141a** and is then cured in the bobbin mounting seat **146**, the cured protruding adhesive **g** may further restrict the range within which the second stopper **112** is movable, thus causing operational malfunctions of the lens moving apparatus.

Furthermore, when the cured protruding adhesive **g** repeatedly collides with the second stopper **112**, the adhesive **g** may temporarily adhere to the second stopper **112** due to the adhesiveness of the adhesive **g**. Hence, the cured protruding adhesive **g** may temporarily restrict the movement of the second stopper **112**, thus causing operational malfunctions of the bobbin **110** including the second stopper **112** or of the entire lens moving apparatus.

As shown in FIG. **16**, the second flow restrictor **146a-1** may be embodied as an inclined surface for restricting the flow of the adhesive **g**. Specifically, in the case of a sharp edge having no second flow restrictor **146a-1**, when the adhesive **g** flows down from the first magnet mounting seat **141a** and reaches the lower surface **146a** of the bobbin mounting seat **146**, the adhesive **g** may partially infiltrate into the bobbin mounting seat **146**. When the bobbin **110** is first coupled to the housing **140** during the operation of assembling the lens moving apparatus, a portion of the adhesive **g** may drop on the second stopper **112** of the bobbin **110** and may be cured thereon.

Even when the adhesive **g** does not drop, a protruding portion may be created on the end of the lower surface **146a** of the bobbin mounting seat **146** in the first direction, and

such a protruding portion may also cause operational malfunctions of the second stopper **112**.

In contrast, in the case in which the second flow restrictor **146a-1** is embodied as an inclined surface, when the adhesive **g** flows down and reaches the upper end of the second flow restrictor **146a-1**, the adhesive **g** flows along the inclined surface while changing its flowing direction. Accordingly, since the adhesive **g** adheres to the inclined surface, the adhesive **g** does not flow down any further, and does not create a protruding portion.

As the width **w4** of the second flow restrictor **146a-1**, that is, the width of the inclined surface in the cross-sectional view of FIGS. **15** and **16** is increased, the effect of restricting the flow of the adhesive **g** may be increased. Accordingly, the width **w3** of the second flow restrictor **146a-1** is preferably as large as possible, as long as it has no influence on the overall operation of the lens moving apparatus.

FIG. **17** is a view showing the adhesive inlet **141b** according to one embodiment. FIG. **18** is a view showing the adhesive inlet **141b** according to another embodiment.

As shown in FIG. **17**, the adhesive inlet **141b** may be formed in the first side surface **P1-1**. In this case, when additional adhesive **g** is injected into the bonding area from the injector **n**, the adhesive **g**, discharged from the point of the injector, first permeates the bonding area on the first side surface **P1-1** of the first magnet mounting seat **141a**, and then permeates the second side surface **P1-2** and the upper surface **P2** of the first magnet mounting seat **141a**.

In contrast, as shown in FIG. **18**, the second side surfaces **P1-2** may be provided with the respective adhesive inlets **141b**. In this case, although one of the two second side surfaces **P1-2** may be provided with the adhesive inlet **141b**, both of the second side surfaces **P1-2** are preferably provided with respective adhesive inlets **141b**, in order to allow the adhesive **g** to be uniformly applied to both of the side surfaces **P1-2**.

In this case, when additional adhesive **g** is injected to the bonding area from the injector **n**, the adhesive **g** discharged from the point of the injector first permeates the bonding area on the second side surface **P1-2** of the magnet mounting seat **141a** and then permeates the first side surface **P1-1** and the upper surface **P2** of the first magnet mounting seat **141a**.

Alternatively, the embodiments shown in FIGS. **17** and **18** may be combined with each other. In other words, the adhesive inlet **141b** may be formed in both the first side surface **P1-1** and the second side surface **P1-2**. In this case, the adhesive inlets **141b** are preferably formed in both of the second side surfaces **P1-2**, respectively, in order to allow the adhesive **g** to be uniformly applied to both of the second side surfaces **P1-2**. Consequently, a total of three adhesive inlets **141b** may be provided in this embodiment.

FIG. **19** is a view illustrating the second flow restrictor **146a-1** according to another embodiment. As shown in FIG. **19**, a plurality of stepped portions **s** may be formed at portions where the inner surface of the housing **140** meets the bobbin mounting seat **146**, and the individual stepped portions **s** may be provided with respective second flow restrictors **146a-1**.

In other words, a plurality of second flow restrictors **146a-1** may be provided. The embodiment including the plurality of second flow restrictors **146a-1** may be more effective in restricting the flow of the adhesive **g** by virtue of the second flow restrictors **146a-1**, compared to the embodiment including only one second flow restrictor **146a-1**.

The plurality of flow restrictors **146a-1** may be formed, for example, in such a manner as to form a plurality of stepped portions **s** at areas where the inner surface of the

housing meets the upper surface **P2** of the first magnet mounting seat **141a** and the lower surface **146a** of the bobbin mounting seat **146** and to chamfer the respective stepped portions **s**.

Although two second flow restrictors **146a-1** are provided in this embodiment, the disclosure is not limited thereto. The number of second flow restrictors **146a-1** may be appropriately selected in consideration of the available space in the housing **140** and influences on the overall operation of the lens moving apparatus.

The second flow restrictors **146a-1** may be provided at all or some of the plurality of stepped portions **s**. In the case where the second flow restrictors **146a-1** are provided at only some of the plurality of stepped portions **s**, the first portion where the adhesive **g**, which has flowed from the upper surface **P2**, is introduced into the bobbin mounting seat **146**, that is, the first stepped portion **s** which is directly connected to the inner surface of the housing **140**, is necessarily provided with the second flow restrictor **146a-1**.

Alternatively, although the plurality of stepped portions **s** are provided, the plurality of stepped portions **s** may not be chamfered. In this case, the plurality of stepped portions **s** may serve as the second flow restrictors.

In the embodiment, the second flow restrictors **146a-1** serve to prevent the adhesive **g**, which has flowed from the first magnet mounting seat **141a**, from being introduced into the bobbin mounting seat **146** and being cured thereat, or to serve to prevent the adhesive **g** from being cured at the lower end **146a** of the bobbin mounting seat **146** and protruding therefrom, thus preventing operational malfunctions of the overall lens moving apparatus.

FIG. **20** is a schematic perspective view showing the lens moving apparatus according to another embodiment. FIG. **21** is an enlarged view showing area **A** of FIG. **20**.

As shown in FIGS. **20** and **21**, the lens moving apparatus according to the embodiment may include the printed circuit board **250**, the cover member **300**, and a recess **g1** defined by a portion of the base **210** surrounding the recess **g1**.

The printed circuit board **250** may include at least one bent terminal member **253**, and the first recess **g1** may be formed in both sides of the upper portion of the terminal member **253**. The both sides of the upper portion of the terminal member **253** may be provided with recesses **253a** constituting a part of the first recess **g1**.

The terminal member **253** may be provided with a plurality of terminals **251** so as to receive external power and supply the power to the first coil **120** and the second coil **230**. The number of terminal members **253** may be increased or decreased depending on the kinds of components to be controlled.

The first recess **g1** is intended to allow an adhesive to be introduced therethrough so that the adhesive penetrates the gap between the cover member **300** and the base **210** and fills the gap.

The first recess **g1** may be surrounded by a first surface **f1**, which is defined by one end of the terminal member **253**, a second surface **f2**, which is defined by one end of the cover member **300**, and a third surface **f3**, which is defined by one end of the base **210**. The first recess **g1** may be open at the front side and may be closed by the outer surface **210a** of the base **210**.

As shown in FIG. **21**, the outer surface **210a** of the base **210**, which defines the first recess **g1**, may be provided at the center thereof with a projection. Here, since the gaps between the printed circuit board **250**, the cover member **300** and the base **210** face the side surface of the projection, adhesive that has been introduced into the first recess **g1**

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flows toward the end of the projection and thus easily penetrates the gaps, thus making it easy to perform bonding using an adhesive.

The first recess **g1** may include a pair of first recesses, which are spaced apart from each other by a predetermined distance. Since the gaps between the printed circuit board **250**, the cover member **300** and the base **210** are positioned at both sides of an upper portion of the terminal member **253**, the pair of first recesses **g1** are provided in order to bond the gaps.

The adhesive that is used in the bonding procedure may be made of polymer-based material such as epoxy. When the adhesive has a low viscosity, although the bonding procedure of filling the gap may be easily performed by virtue of the high fluidity, the bonding force may be decreased. On the other hand, when the adhesive has a high viscosity, although the adhesive has a high bonding force, the bonding procedure may not be easily performed due to the low fluidity.

Therefore, considering facilitation of work, the bonding strength between the printed circuit board **250**, the cover member **300** and the base **210** after the bonding procedure, there is a need to perform the bonding procedure using an adhesive having an appropriate viscosity.

The width **w** of the first recess **g1**, which is measure in the length direction of the terminal member **253**, may be determined in consideration of whether the needle of the adhesive injector, which is used in the bonding procedure, can easily enter the first recess **g1**, ease of formation of the first recess **g1**, facilitation of the bonding procedure, and the like.

Considering these matters, the width **w** of the first recess **g1** is 0.1 mm or more, preferably 0.3 mm or more, and more preferably 0.1 mm-0.6 mm.

FIG. **22** is a view showing a bonding procedure of coupling the lens moving apparatus according to another embodiment.

The bonding procedure of coupling the terminal member **253**, the cover member **300** and the base **210**, which are provided at the printed circuit board **250**, may be performed using an application device on which the adhesive injector is mounted.

As shown in FIG. **22**, the bonding procedure may be performed, for example, by filling and bonding, using the adhesive, a first gap **G1** between the terminal member **253** and the cover member **300** and a second gap **G2** between the terminal member **253**, the cover member **300** and the base **210** while moving the needle of the adhesive injector in the length direction of the terminal member **253**, that is, in the horizontal direction in the drawing.

As shown in FIG. **22**, when the bonding procedure is performed in the lens moving apparatus equipped with the first recess **g1** according to the embodiment while moving the needle from the left to the right, the bonding procedure may be easily implemented at the first gap **G1**. Furthermore, since the second gap **G2** is provided with the first recess **g1**, the bonding procedure may also be easily implemented at the second gap **G2**. The specific bonding procedure will now be described hereinafter.

The needle is first disposed at the first recess **g1** positioned at the second gap **G2**, which is represented at the left side of the drawing, and sufficient adhesive is injected into the first recess **g1**. The injected adhesive is introduced into the first recess **g1**, and is further introduced into the second gap **G2**, where the terminal member **253**, the cover member **300** and the base **210** define the first recess **g1**.

A sufficient amount of adhesive is introduced into the second gap **G2** through the first recess **g1**, and thus the terminal member **253**, the cover member **300** and the base

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210 are bonded to each other with a sufficient coupling force, and the second gap **G2** may be completely sealed. After the bonding procedure at the second gap **G2** is completed, a procedure of sealing the first gap **G1** may be performed.

Subsequently, a bonding procedure is performed in such a manner as to fill the first gap **G1** with an adhesive while moving the needle positioned at the first recess **g1** to the right in the drawing.

At this point, by injecting a sufficient amount of adhesive into the first gap **G1** and appropriately controlling the moving speed of the needle, the terminal member **253** and the cover member **300** may be coupled to each other with a sufficient coupling force therebetween, and the first gap **G1** may be completely sealed.

Subsequently, the needle is disposed at the first recess **g1**, which is positioned at the second gap **G2**, which is represented at the right of the drawing, and the same bonding procedure as that performed at the second gap **G2**, which is represented at the left of the drawing, is performed.

Through these procedure, the terminal member **253**, the cover member **300** and the base **210** may be bonded to each other with a sufficient coupling force therebetween, and the second gap **G2** may thus be completely sealed.

According to the embodiment, since the bonding procedure is performed by injecting the adhesive into the first recess **g1**, the gap between the terminal member **253**, the cover member **300** and the base **210** is completely sealed, and the terminal member **253**, the cover member **300** and the base **210** may thus be coupled to each other with a sufficient coupling force therebetween.

FIG. **23** is a bottom view showing the base **210** according to one embodiment. The base **210** according to the embodiment may include a second recess **g2**, which is positioned between the bottom surface thereof and the cover member **300** or the printed circuit board **250** so as to bond the base **210** to the cover member **300** or the printed circuit board **250**.

At this point, the terminal member **253**, which is bent from the printed circuit board **250**, and the base **210** are bonded to each other, and the second recess **g2** may be provided at the area where the base **210** is coupled to the terminal member **253**. Meanwhile, the cover member **300** and the base **210** are bonded to each other, and the second recess **g2** may be provided at the area where the cover member **300** is coupled to the base **210**.

In order to block the infiltration of external pollutants into the inside of the lens moving apparatus, there is a need to perform a bonding procedure. However, in the case where the gap between the terminal member **253**, the base **210** and the cover member **300** is sealed with an adhesive, the bonding procedure is not so easy due to the presence of the terminals of the terminal member **253**.

Specifically, when the bonding procedure is performed at the front surface, at which the terminals of the terminal member **253** are formed, the adhesive may adhere to the terminals. After the assembly of the lens moving apparatus including the bonding procedure, the adhesive adhering to the terminals may cause bad connections, malfunctions attributable to the bad connections, or the like during tests of the performance of the lens moving apparatus or upon actual use of the lens moving apparatus.

When the bonding procedure is performed at the front surface of the terminal member **253** at which the terminals are formed, since a plurality of projections or recesses are present at the area where the terminal member **253**, the base **210** and the cover member **300**, which may obstruct the movement of the needle for applying the adhesive, are

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coupled to each other, the bonding procedure may not be smoothly implemented, and the needle may be bent or broken during the bonding procedure.

Accordingly, in the embodiment, the bonding procedure is performed at the rear surface of the terminal member 253 rather than at the front surface so as to make it easy to implement the bonding between the terminal member 253, the base 210 and the cover member 300 and to achieve complete sealing by virtue of the bonding procedure.

The base 210 may be configured to have a square contour, and the second recesses g2 may be provided at respective sides of the square shape. The area where the cover member 300 or the terminal member 253 of the printed circuit board 250 is coupled to the base 210 may be provided at the respective sides of the square shape.

The second recess g2 may be formed close to the gap between the terminal member 253, the base 210 and the cover member 300, and may be closed at the inner side and open at the outer side so as to communicate with the gap.

Consequently, the adhesive is injected into the second recess g2, and is subsequently introduced into the gap positioned at the open area of the second recess g2 so as to fill the gap, thus completing the bonding procedure. By injecting an appropriate but sufficient amount of adhesive into the second recess g2, the gap may be completely filled and sealed with the adhesive.

FIG. 24 is a bottom view showing a base 210 according to another embodiment. As shown in FIG. 24, the second recess g2 may be provided with a third recess g3, which is recessed toward the center of the base 210.

The third recess g3 may be additionally provided at the second recess g2. By virtue of the provision of the third recess g3, when an adhesive is injected into the third recess g3, the adhesive may be introduced into the gap through the second recess g2 from the third recess g3, thus sealing the gap.

A point application technology of injecting an adhesive into the third recess g3 at a single point may have effects of remarkably reducing breakage and bending of the needle compared to a line application technology of disposing the needle at an initial position and moving the needle along the gap.

FIG. 25 is a bottom view showing the base 210 according to a further embodiment. As shown in FIG. 25, the second recess g2 may be provided with a fourth recess g4, which is recessed toward the corner of the base 210.

The fourth recess g4 may include a plurality of fourth recesses g4, which are symmetrically positioned with respect to an imaginary center line that extends through the center of each side of the base 210 and the center of the base 210. Although FIG. 25 illustrates a pair of fourth recesses g4 at each fourth recess g4, the disclosure is not limited thereto.

Specifically, the fourth recess g4 may include four or more fourth recesses g4, which are symmetrically positioned. Although the corresponding fourth recesses g4 are symmetrically positioned with respect to each other, there is no need to configure the corresponding fourth recesses g4 to have symmetrical shapes.

Like the third recess g3, the fourth recess g4 may be additionally provided at the second recess g2. When an adhesive is injected into the fourth recess g4 during the bonding procedure, the adhesive may be introduced into the gap through the second recess g2 from the fourth recess g4, thus sealing the gap.

As in the preceding example, the point application technology of injecting an adhesive into the third recess g4 at a single point may have effects of remarkably reducing break-

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age and bending of the needle compared to the line application technology of disposing the needle at an initial position and moving the needle along the gap.

FIG. 26 a bottom view showing a base 210 according to still another embodiment. As shown in FIG. 26, the fourth recesses g4 are formed at respective corners of the base 210, and two adjacent fourth recesses g2, with the corner disposed therebetween, may be connected to each other via the associated fourth recess g4.

Accordingly, the second recesses g2, which are formed in the bottom surface of the base 210, may be connected to each other via the fourth recesses g4. Here, an adhesive, which has been injected into the fourth recesses g4, may be introduced into the second recesses g2, which are adjacent to each other with the corners of the base 210 therebetween, and may be introduced into the gaps, which are adjacent to each other with the corners of the base 210 therebetween.

FIG. 27 is a bottom view showing a base 210 according to yet another embodiment. As shown in FIG. 27, the second recess g2 may be provided with the third recess g3, which is recessed toward the center of the base 210, and the fourth recess g4, which is recessed toward the corner of the base 210. The fourth recess g4 may include a plurality of fourth recesses g4, which are symmetrically positioned with respect to each other.

In this embodiment, the point application technology may be performed so as to inject an adhesive into the third recess g3 and/or the fourth recess g4. Like the embodiment shown in FIG. 26, the fourth recesses g4 may be formed at the corners of the base 210, and the second recesses g2, which are adjacent to each other with the corners of the base 210 therebetween, may be connected through the fourth recesses g4.

FIG. 28 is a view showing the second recess g2. FIG. 29 is a schematic view showing a side surface f5 of the second recess g2.

As shown in FIGS. 28 and 29, the second recess g2 including the fourth recess g4 may be inclined toward the end from the center of the side of the base 210 so as to be increasingly widened toward the adjacent side of the square base 210.

Thanks to the inclined structure, adhesive p, which has been introduced into the fourth recess g4, may easily flow into the second recess g2 so as to fill the gap, that is, the area to be bonded, thus enable the bonding procedure to be rapidly completed.

As shown in FIG. 29, the fourth recess g4 may be configured such that the minimum depth L1 of the fourth recess g4 is greater than the maximum depth L2 of the second recess g2. During the bonding procedure, the upper end of the side surface f5 of the second recess g2 including the fourth recess g4 is positioned to be horizontal or almost horizontal. Accordingly, the side surface f5 of the second recess g2 is inclined, and the bottom of the second recess g2 is also inclined.

Consequently, the adhesive p, which has been introduced into the fourth recess g4, may smoothly flow along the bottom of the second recess g2, and may then be introduced in the gap, that is, the area to be bonded, thus sealing the gap.

Any of the second recess g2, the third recess g3 and the fourth recess g4 in the respective embodiments shown in FIGS. 23 to 27 may be inclined toward the outside from the inside of the base 210 so as to be increasingly deepened.

Accordingly, since the respective recesses formed in the bottom surface of the base 210 are deepened toward the outside from the inside of the base 210, adhesive p intro-

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duced in the respective recesses may more smoothly flow into the gaps, thus enabling the bonding procedure to be rapidly completed.

The lens moving apparatus according to the above embodiments may be applied to products in various fields, for example camera modules. Such a camera module may be applied, for example, to mobile devices such as cellular phones.

The camera module according to the embodiment may include a lens barrel coupled to the bobbin **110**, an image sensor (not shown), a printed circuit board **250** and an optical system.

The lens barrel is as described above, and the printed circuit board **250** is the component on which the image sensor is mounted, and may constitute the bottom surface of the camera module.

The optical system may include at least one lens for transmitting an image to the image sensor. The optical system may be provided with an actuator module capable of fulfilling functions of autofocus and handshake correction. The actuator module functioning to fulfill the function of autofocus may be variously constructed, and a voice coil unit motor is primarily used in the actuator module. The lens moving apparatus according to the embodiments may serve as an actuator module fulfilling both functions of autofocus and handshake correction.

The camera module may further include an infrared ray screening filter (not shown). The infrared ray screening filter functions to block the incidence of light in the infrared range on the image sensor. In the base **210** illustrated in FIG. 2, the infrared ray screening filter may be installed at a position corresponding to the image sensor, and may be coupled to a holder member (not shown). The base **210** may support the lower portion of the holder member.

The base **210** may be provided with an additional terminal member for conductive connection to the printed circuit board **250**, and the terminal may be integrally formed using a surface electrode. The base **210** may serve as a sensor holder for protecting the image sensor. In this case, although a projecting portion may be formed downward along the lateral side surface of the base **210**, it is not an essential component. Although not shown in the drawings, an additional sensor holder may be disposed under the base **210** to serve as the projecting portion.

Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

What is claimed is:

1. A lens moving apparatus comprising:

a base;

a circuit board disposed on the base and comprising a terminal portion disposed on an outer side surface of the base;

a bobbin disposed above the base and configured to move in an optical axis direction;

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a cover member comprising an upper plate and a side plate connecting the upper plate and configured to accommodate the bobbin and at least a portion of the circuit board; and

a first adhesive bonding the base, the circuit board, and the cover member,

wherein the terminal portion of the circuit board comprises a curved surface formed at a side of an upper portion thereof, and the curved surface is recessed away from a portion of the outer side surface of the base,

wherein a first recess is formed and is surrounded by the side plate of the cover member, the curved surface of the terminal portion of the circuit board, and the portion of the outer side surface of the base, and

wherein the base comprises a protrusion formed on the portion of the outer side surface of the base, and the first adhesive is disposed in the first recess.

2. The lens moving apparatus according to claim 1, wherein a first gap exists between the terminal portion of the circuit board and the protrusion of the base, and a portion of the first adhesive is disposed in the first gap.

3. The lens moving apparatus according to claim 1, wherein a second gap exists between the side plate of the cover member and the protrusion of the base, and a portion of the first adhesive is disposed in the second gap.

4. The lens moving apparatus according to claim 1, wherein a third gap exists between the end of the base and the protrusion of the base, and a portion of the first adhesive is disposed in the third gap.

5. The lens moving apparatus according to claim 1, wherein a first gap exists between the terminal portion of the circuit board and the protrusion of the base, and wherein a second gap exists between the side plate of the cover member and the protrusion of base, and a third gap exists between the end of the base and the protrusion of the base, and a portion of the first adhesive is disposed in the third gap, and

wherein the first adhesive is disposed in the first to third gaps.

6. The lens moving apparatus according to claim 1, wherein the first recess is formed at each of both sides of the terminal portion of the circuit board.

7. The lens moving apparatus according to claim 1, wherein the first recess is open at a front side thereof and closed at a rear side thereof by an outer side surface of the base.

8. The lens moving apparatus according to claim 1, wherein the terminal portion of the circuit board comprises a plurality of terminals.

9. The lens moving apparatus according to claim 1, wherein the base comprises a second recess in a portion of a bottom surface thereof for applying a second adhesive for bonding the circuit board to the base.

10. The lens moving apparatus according to claim 9, wherein the base has a square shape contour and the second recess is provided at each side of the bottom surface of the base.

11. The lens moving apparatus according to claim 1, wherein the base comprises a support recess formed in the outer side surface thereof, and the terminal portion of the circuit board is disposed in the support recess.

12. The lens moving apparatus according to claim 1, comprising:

a housing disposed above the circuit board and configured to accommodate the bobbin;

a magnet disposed on the housing;

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a first coil disposed on the bobbin and configured to move the bobbin in the optical axis direction by an interaction with the magnet.

13. The lens moving apparatus according to claim 12, comprising:

an upper elastic member coupled to an upper portion of the bobbin and an upper portion of the housing;
a second coil configured to move the housing by an interaction with the magnet; and
a support member electrically connecting the upper elastic member and the circuit board.

14. The lens moving apparatus according to claim 13, wherein the second coil is electrically connected to the circuit board.

15. A lens moving apparatus comprising:

a base;

a circuit board disposed on the base and comprising a terminal portion disposed on an outer side surface of the base and a terminal disposed on the terminal portion;

a bobbin disposed above the circuit board and configured to move in an optical axis direction;

a cover member comprising an upper plate and a side plate connecting the upper plate and configured to accommodate the bobbin and at least a portion of the circuit board; and

a first adhesive bonding the base, the circuit board, and the cover member,

wherein the terminal portion of the circuit board comprises a curved surface formed at a side of an upper portion thereof, and the curved surface is recessed away from a portion of the outer side surface of the base,

wherein a first recess is formed and is surrounded by one end of the side plate of the cover member, the curved surface of the terminal portion of the circuit board, and the portion of the outer side surface of the base,

wherein the base comprises a protrusion formed on the portion of the outer side surface of the base,

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wherein a first gap is formed between the one end of the terminal portion of the circuit board and the protrusion of the base, and

wherein a first portion of the first adhesive is disposed in the first gap.

16. The lens moving apparatus according to claim 15, wherein a second gap is formed between the one end of the side plate of the cover member and the protrusion of the base, and

wherein a second portion of the first adhesive is disposed in the second gap.

17. The lens moving apparatus according to claim 15, wherein a third gap is formed between the one end of the base and the protrusion of the base, and

wherein a third portion of the first adhesive is disposed in the third gap.

18. The lens moving apparatus according to claim 15, comprising:

a housing disposed above the circuit board and configured to accommodate the bobbin;

a magnet disposed on the housing;

a first coil disposed on the bobbin and configured to move the bobbin in the optical axis direction by an interaction with the magnet;

an upper elastic member coupled to an upper portion of the bobbin and an upper portion of the housing;

a second coil configured to move the housing by an interaction with the magnet; and

a support member electrically connecting the upper elastic member and the circuit board.

19. The lens moving apparatus according to claim 15, wherein the base comprises a second recess formed in a portion of a bottom surface of the base, and the second recess is for applying a second adhesive for bonding the circuit board to base.

20. A camera module comprising:

a lens;

the lens moving apparatus according to claim 1; and
an image sensor.

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